PROJECT PRESENTATION

Computational Linguistics 1

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What we did

01

Manually Annotating Hindi and English Datasets
200 sentences each

02

Changed the tagging scheme based on the research paper. Manually annotated 50 english and 50 hindi sentences with that.

03

Ran a python script to generaliza and tag all 200 tags for eng with new scheme and made confusion matrix.

What we did

04

Tested capabilities of different AI models in pannian annotation, using zeroshot, few shot and instruction based methods

05

Annotated Telugu and Gujarati Sentences

06

Paninian tagset to UD conversion

PANINAIN?

Why?
Dependency Parsing

We need to devise a suitable computational grammar formalism for morphologically rich free word order languages.
Paninian karaka roles are semantic, not syntactic. They capture the role a word plays in the action. As this role is independent of word order, this schema holds up well for free word order languages like Hindi, Telugu etc.

Levels in the Paninian Model

```
---  semantic level (what the speaker has in mind)
 |
 |
---  karaka level
 |
 |
---  vibhakti level
 |
 |
---  surface level (uttered sentence)
```

Free Word Order and Vibhakti

```
C.1  raama mohana  ko   piiTataa  hei.
     Ram   Mohan  -ko   beat      is
     (Ram beats Mohan.)

C.2  mohana  ko   raama  piiTataa  hei.
     Mohan -ko    Ram    beat      is
     (Ram beats Mohan.)

C.3  mohana  raama  ko   piiTataa hei.
     Mohan   Ram   -ko   beat     is
     (Mohan beats Ram.)

C.4  raama ko  mohana piiTataa  hei.
     Ram  -ko  Mohan  beat      is
     (Mohan beats Ram.)
```

Why?
Cross Lingual

Hindi and Gujrati: SOV free word order languages with rich morphology
Telugu: Dravidian language with agglutinative morphology
English: SVO and analytical in nature

What we’ll see later: Paninian tags are insufficient for English

PANINIAN FRAMEWORK

1

KĀRAKA: SEMANTICALLY RELATED TO A VERB AS THE DIRECT PARTICIPANTS IN THE ACTION DENOTED BY A VERB ROOT.

2

NON-KĀRAKA: THESE RELATIONS INCLUDE REASON, PURPOSE, POSSESSION, ADJECTIVAL OR ADVERBIAL MODIFICATIONS ETC.

Need for this framework

01 Makes use of primary sources of information

02 Due to the first, it makes grammar economical and easy to write

03 More linguistically elegant as it works based on the principles of that language

TAGSET SHIFT

Old: Karaka Relations

k1: Karta (subject/agent)
k2: Karma (Direct Object/Theme)
k3: Karana (instrument)
k4: Sampradana(recipient/beneficiary)
k5: Apadana(departure/source)
k7: Adhikarana(location/time/context)

Old : Non- Karaka Relations

Standard POS tagging for no-karaka parts of speech:

V:main verb

VAUX: auxiliary verb

JJ: adjective

RB: adverb

PSP: postpositions/case markers

NN: General noun

CC: Conjunction

RP: particle

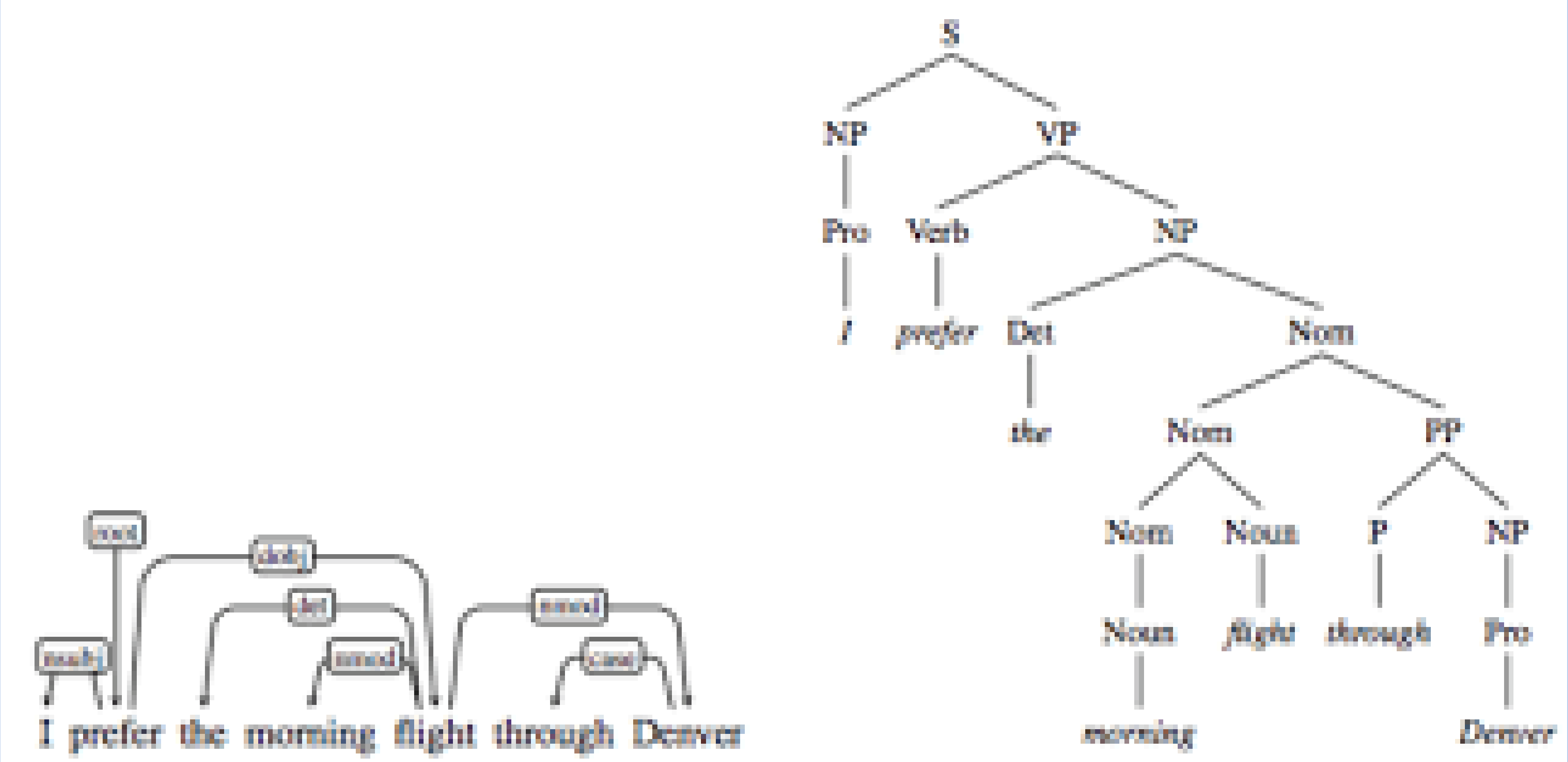
PUNC: punctuation

SYM: for numbers/symbols

New Tagset

Relation	Meaning
k1	Agent / Subject / Doer
k2*	Theme / Patient / Goal
k3	Instrument
k4*	Recipient / Experiencer
k5	Source
k7*	Spatio-temporal
rt	Purpose
rh	Cause
ras	Associative
k*u	Comparative
k*s	(Predicative) Noun / Adjective Complements
r6	Genitives
relc	Modification by Relative Clause
rs	Noun Complements (Appositive)
adv	Verb modifier
adj	Noun modifier

Dependency vs Constituency parsing



Annotations

```
#Sentence 1:
The adj
government k1
announced V
new adj
research adj
grants k2
for -
students k4
. PUNC
```

```
#Sentence 1
Bharat      k1
ne          -
naya        adj
missile     adj
parikshan   k2
saphalta    adj
purvak      adv
kiya        V
.           PUNC
```

```
1   Ahmedabadma - in Ahmedabad
2   varshadni - of rain
3   dhabakedar - heavy/vigorous
4   saruaat - start/beginning
5   thai - happened/occurred
6   che - is
7   . - punc
```

```
1   Pashchimasyalo - west of
2   keela - important
3   parinamam - issue
4   chotushesukundi - occur.pst
5   . - punc
```

Why?

Change the Tagset

Conversion from Pāṇinian Kārakas to Universal Dependencies for Hindi Dependency Treebank

Juhi Tandon, Himani Chaudhary, Riyaz Ahmad Bhat and **Dipti Misra Sharma**
Kohli Center on Intelligent Systems (KCIS),
International Institute of Information Technology,Hyderabad (IIIT-H)
Gachibowli,Hyderabad 500 032, India
{juhi.tandon,himani,riyaz.bhat}@research.iiit.ac.in , dipti@iiit.ac.in

Generalization via Python

```python
def guess_pos(word):
    w = word.lower()
    if w in PUNCTUATION:
        return "PUNC"
    if w in DETERMINERS:
        return "DET"
    if w in PREPOSITIONS:
        return "PREP"
    if w in AUXILIARY_VERBS:
        return "AUX"
```

```python
result = list(sentence)
for i in range(len(result) - 1):
    word, tag   = result[i]
    _, next_tag = result[i + 1]
    if tag in {"k1", "k2"} and next_tag in {"k1", "k2"}:
        result[i] = (word, "adj")
return result
```

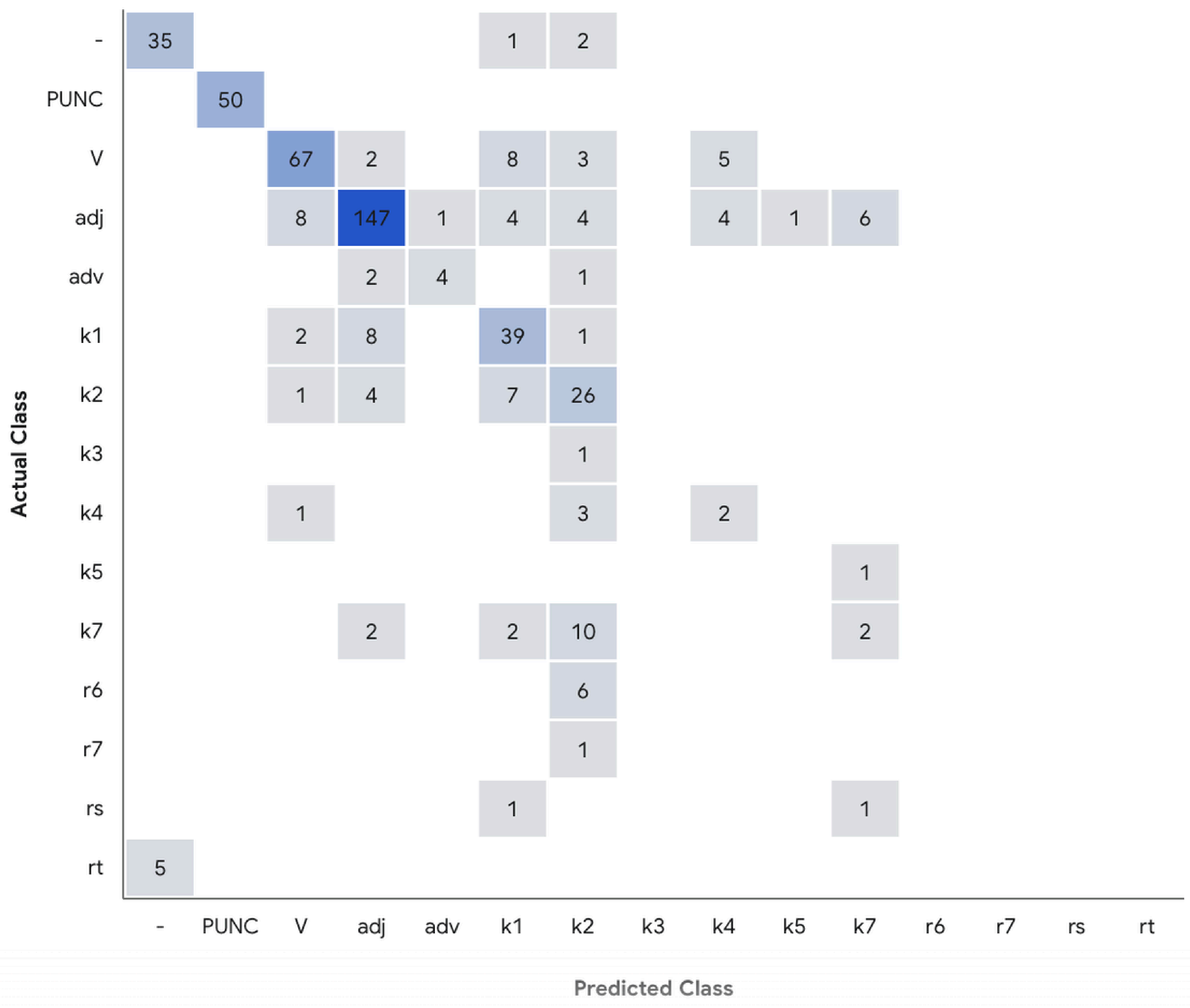
```python
verb_idx = None
for j, w2 in enumerate(words):
    if guess_pos(w2) in {"VERB", "AUX"} and w2.lower() not in AUXILIARY_VERBS:
        verb_idx = j
        break
```

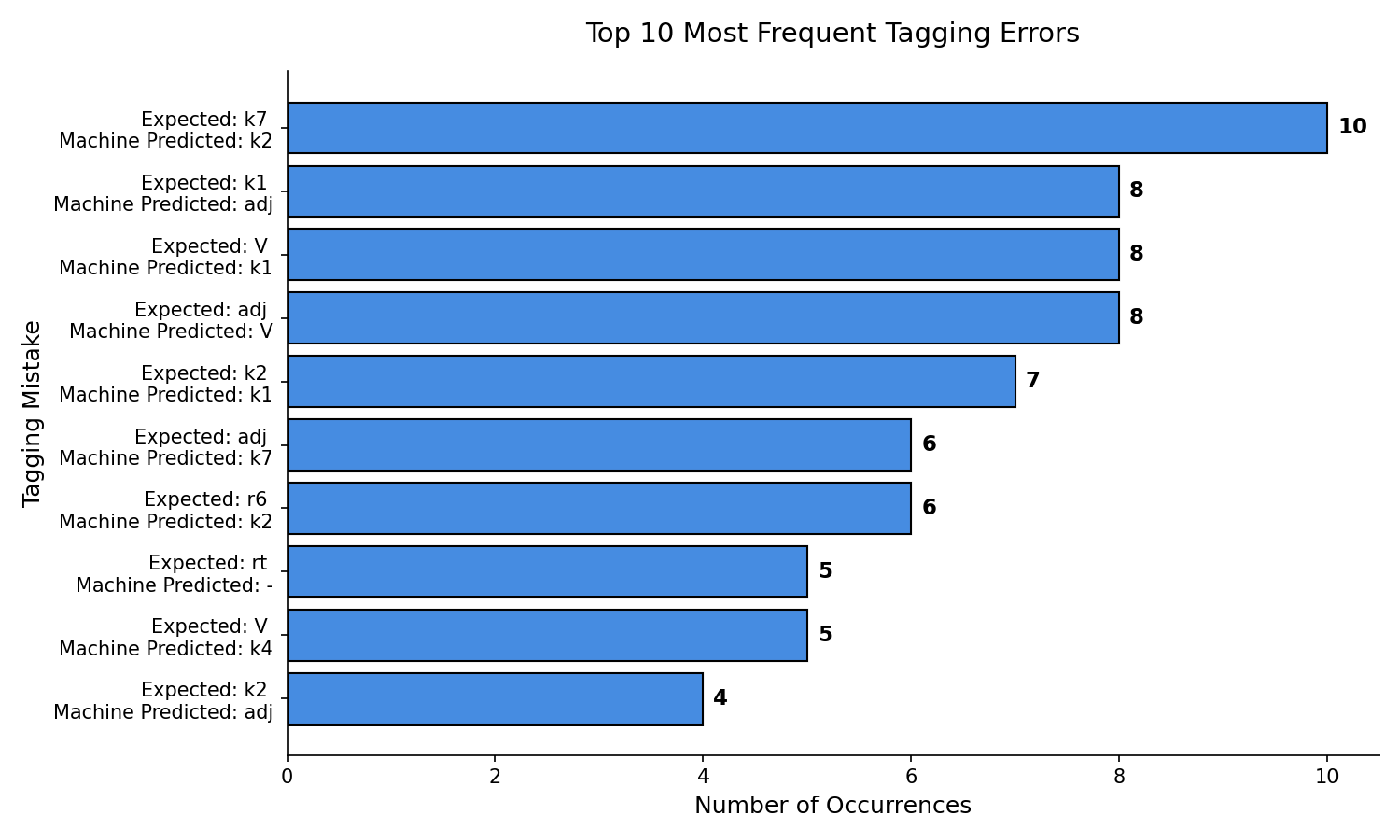
Python based Annotations

```
#Sentence 1:
The adj
government k1
announced V
new adj
research adj
grants k2
for -
students k4
. PUNC
```

```
# The Hindu
# The government announced new research grants for students.
1   The adj
2   government   k1
3   announced    V
4   new adj
5   research     adj
6   grants   k2
7   for -
8   students     k4
9   .   PUNC
```

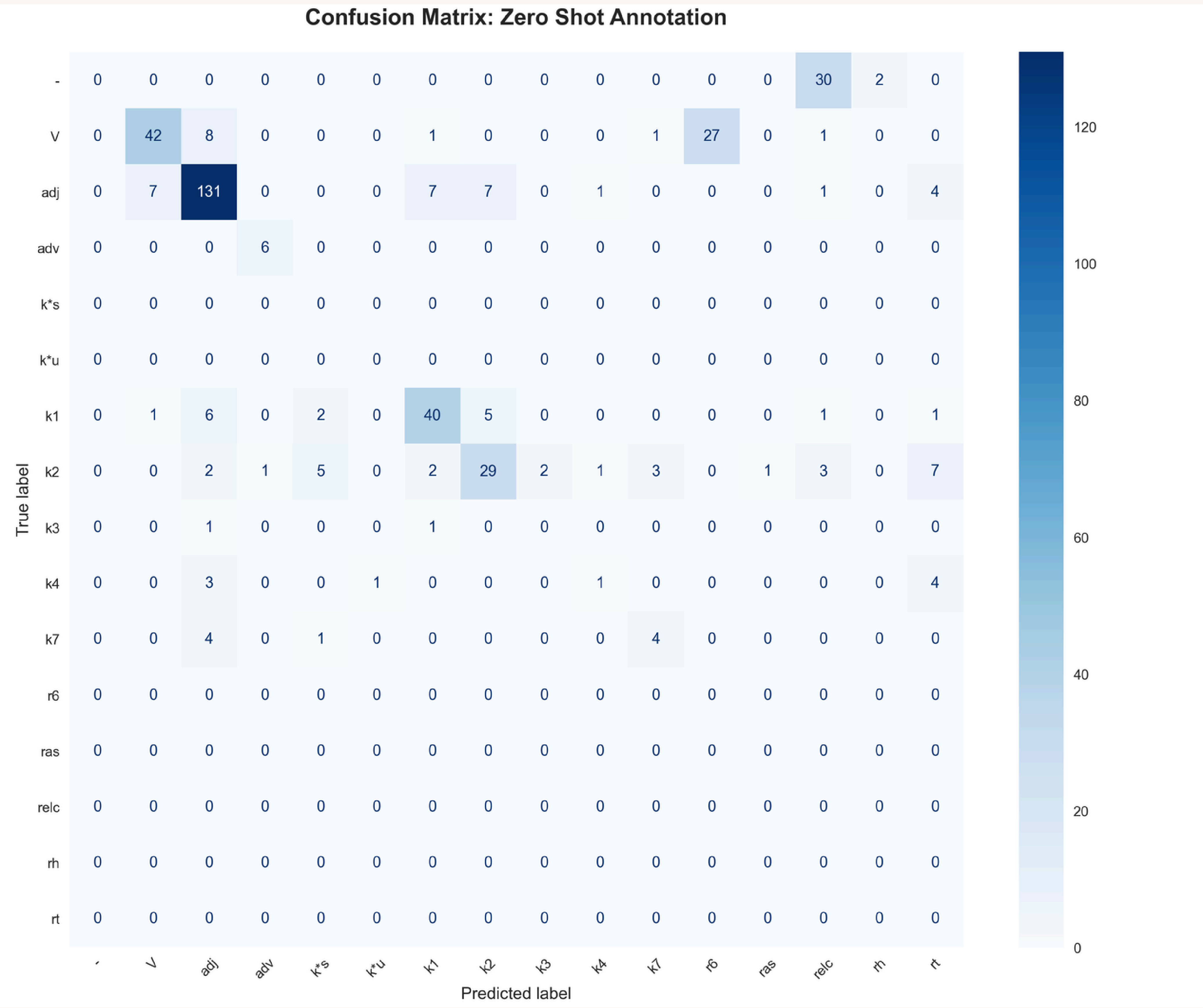
COMPARISIONS

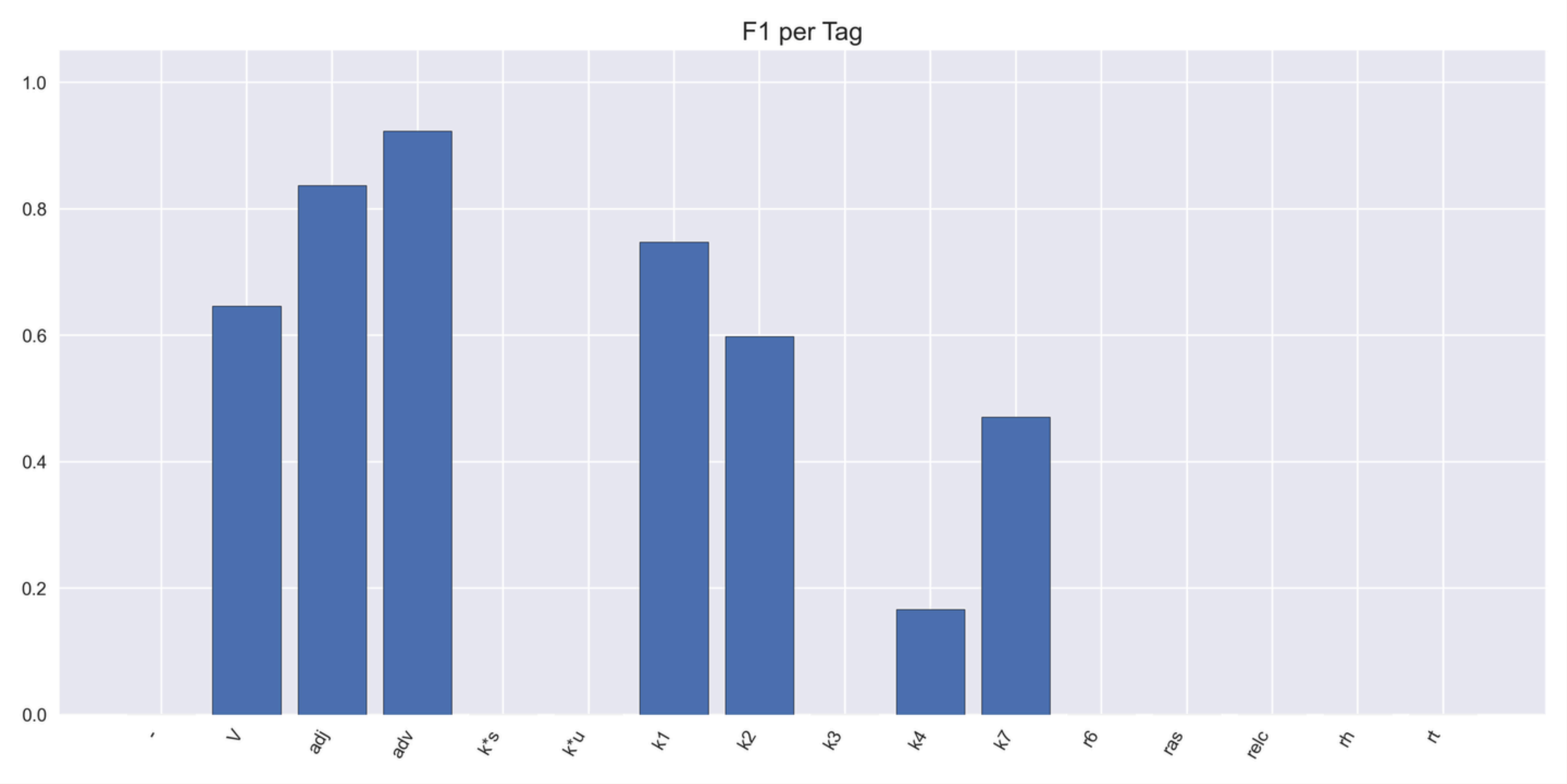

Actual Class
Predicted Class
- PUNC V adj adv k1 k2 k3 k4 k5 k7 r6 r7 rs rt
-: 35, 1, 2
PUNC: 50
V: 67, 2, 8, 3, 5
adj: 8, 147, 1, 4, 4, 4, 1, 6
adv: 2, 4, 1
k1: 2, 8, 39, 1
k2: 1, 4, 7, 26
k3: 1
k4: 1, 3, 2
k5: 1
k7: 2, 2, 10, 2
r6: 6
r7: 1
rs: 1, 1
rt: 5

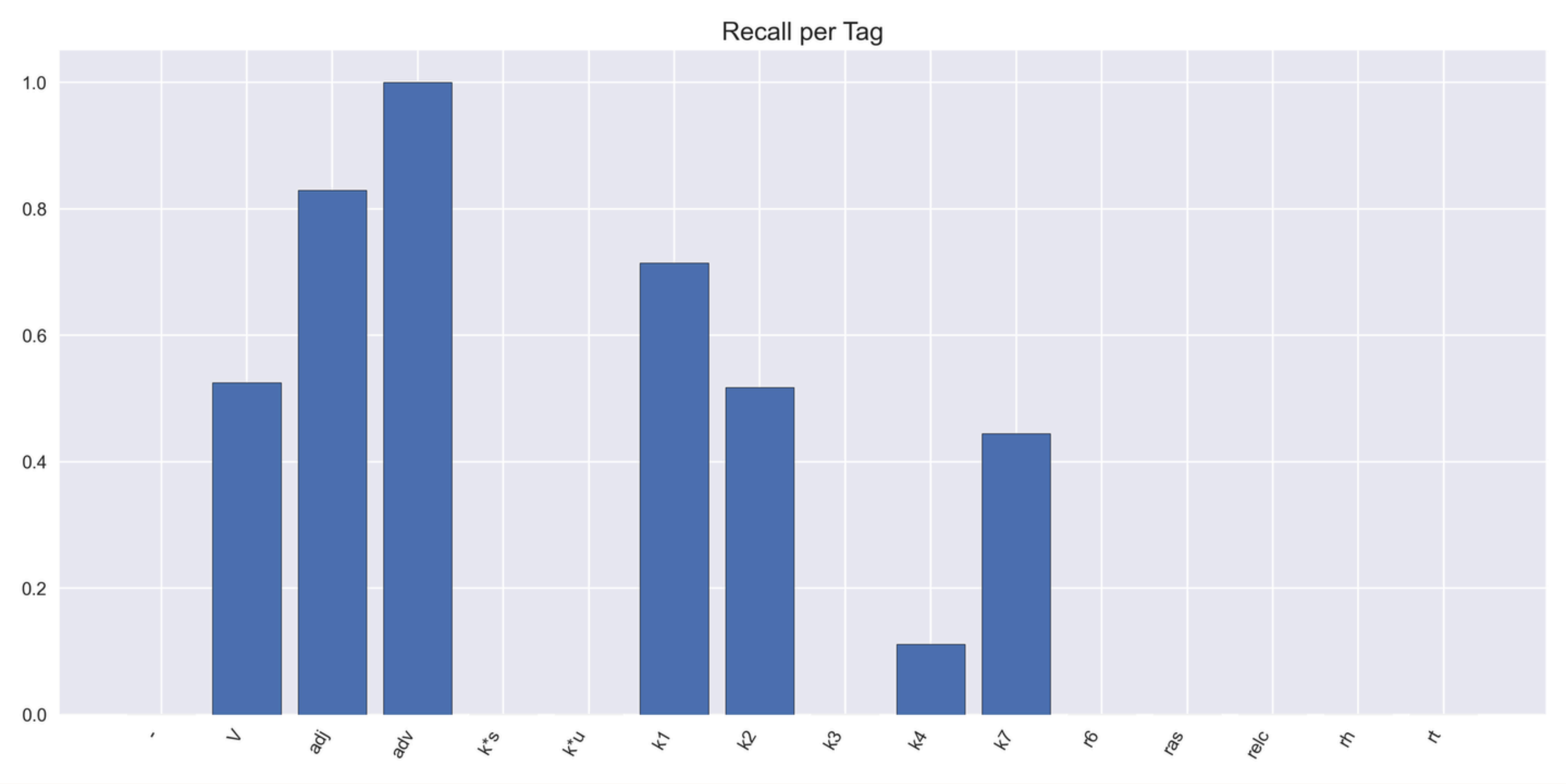

Top 10 Most Frequent Tagging Errors
Tagging Mistake
Expected: k7 Machine Predicted: k2
10
Expected: k1 Machine Predicted: adj
8
Expected: V Machine Predicted: k1
8
Expected: adj Machine Predicted: V
8
Expected: k2 Machine Predicted: k1
7
Expected: adj Machine Predicted: k7
6
Expected: r6 Machine Predicted: k2
6
Expected: rt Machine Predicted: -
5
Expected: V Machine Predicted: k4
5
Expected: k2 Machine Predicted: adj
4
0
2
4
6
8
10
Number of Occurrences

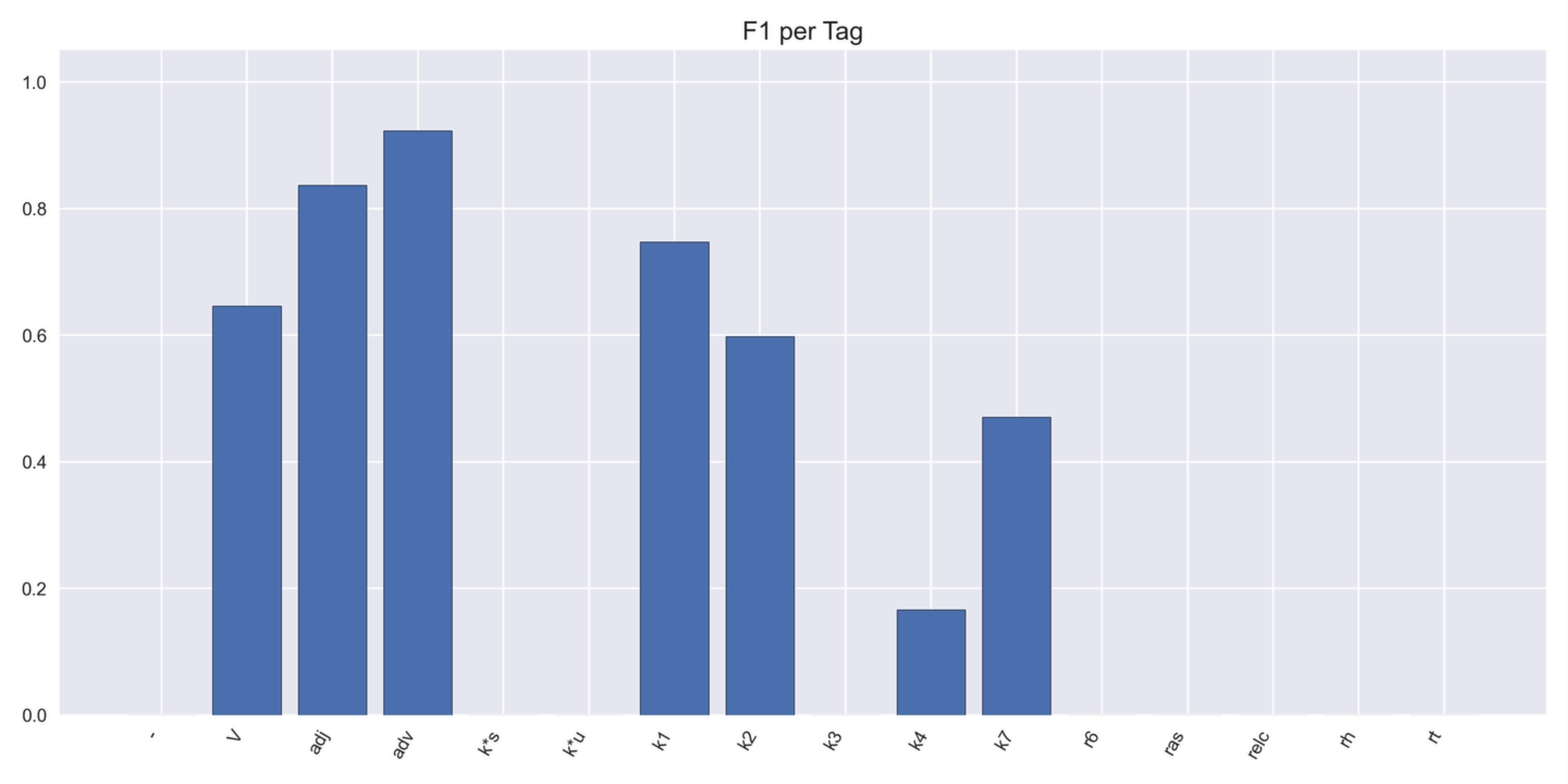
	precision	recall	f1-score	support
-	0.88	0.92	0.9	38
PUNC	1	1	1	50
V	0.85	0.79	0.82	85
adj	0.89	0.84	0.86	175
adv	0.8	0.57	0.67	7
k1	0.63	0.78	0.7	50
k2	0.45	0.68	0.54	38
k3	0	0	0	1
k4	0.18	0.33	0.24	6
k5	0	0	0	1
k7	0.2	0.12	0.15	16
r6	0	0	0	6
r7	0	0	0	1
rs	0	0	0	2
rt	0	0	0	5
accuracy	0.77	0.77	0.77	0.77
weighted avg	0.77	0.77	0.77	481

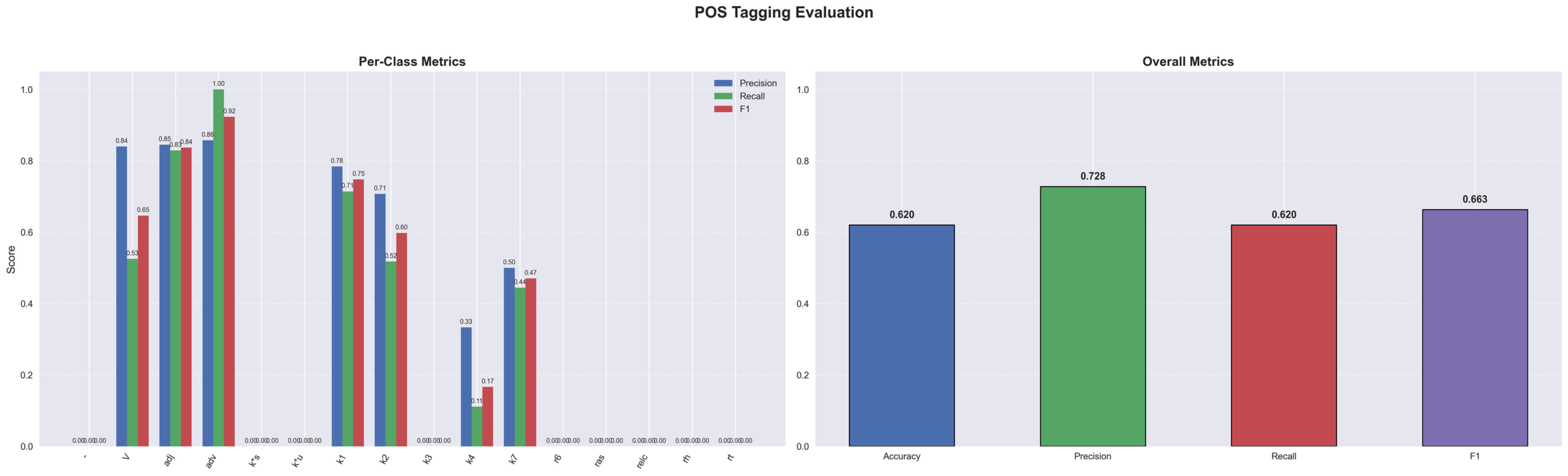
GEMINI: ZERO


Confusion Matrix: Zero Shot Annotation
True label
- V adj adv k*s k*u k1 k2 k3 k4 k7 r6 ras relc rh rt
Predicted label
0 20 40 60 80 100 120


F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k7
r6
ras
relc
rh
rt

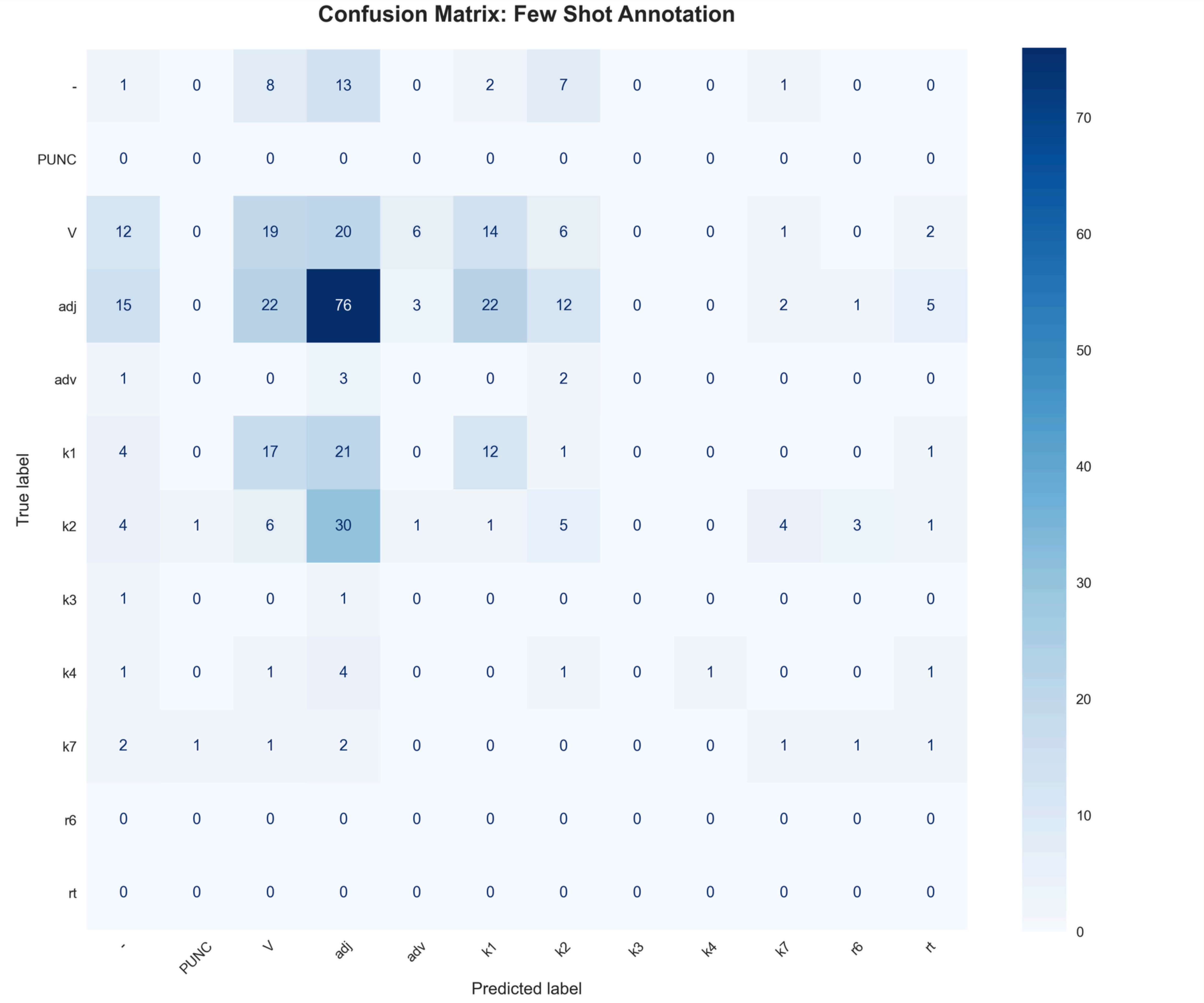

Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k7
r6
ras
relc
rh
rt

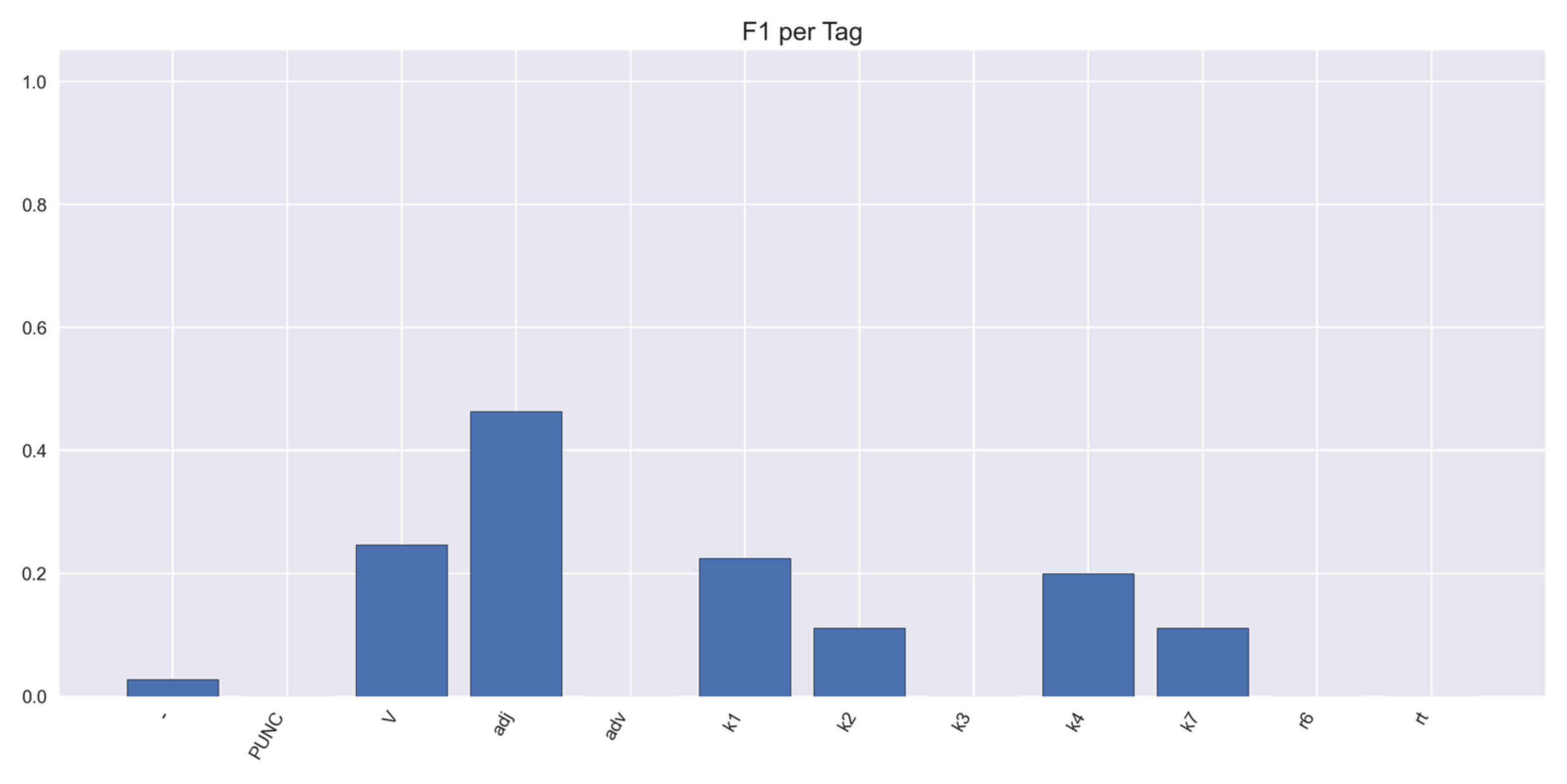

F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k7
r6
ras
relc
rh
rt



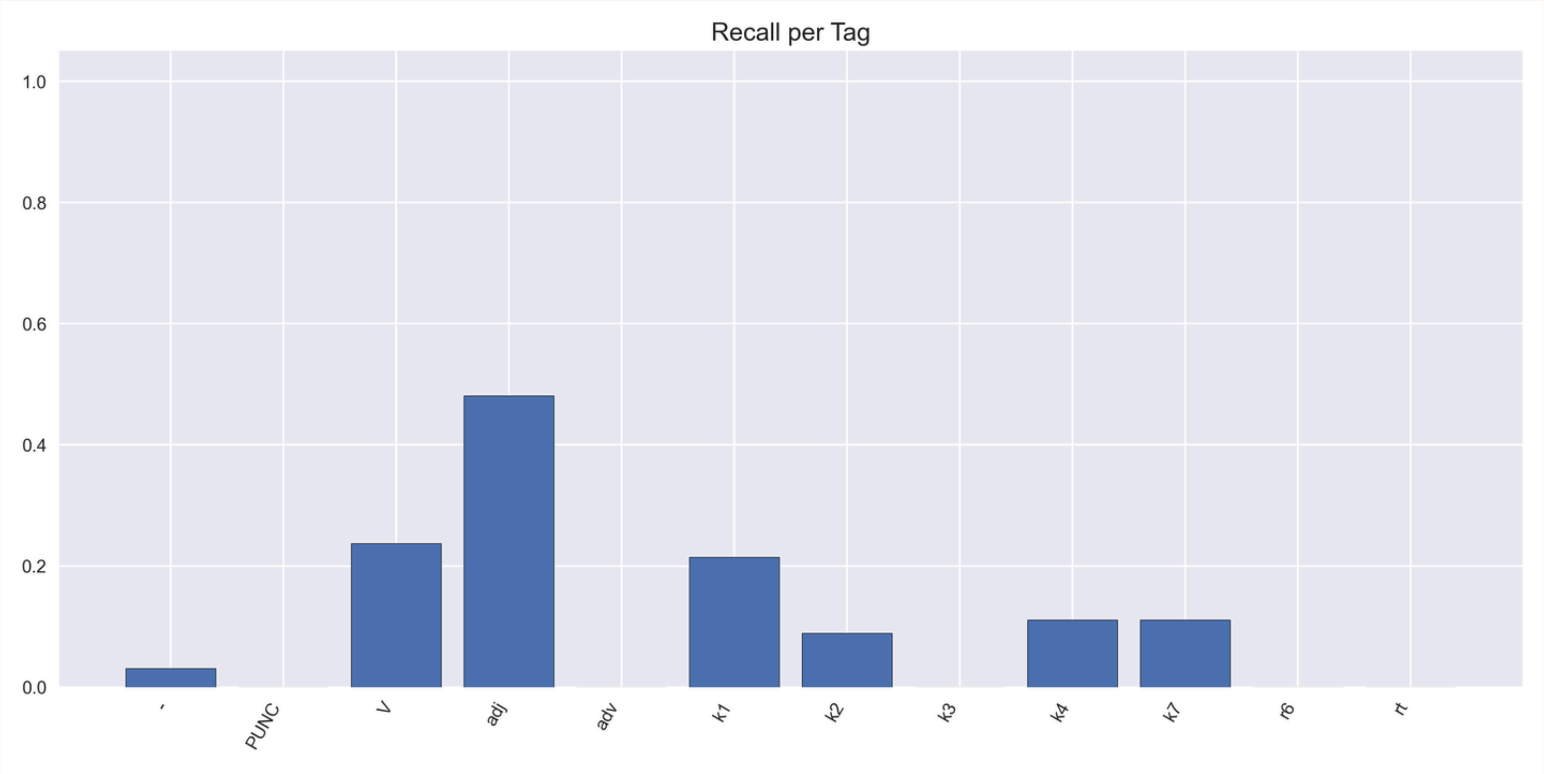
POS Tagging Evaluation
Per-Class Metrics
Precision
Recall
F1
Score
0.84
0.53
0.65
0.85
0.83
0.84
0.86
1.00
0.92
0.78
0.71
0.75
0.71
0.52
0.60
0.33
0.11
0.17
0.50
0.44
0.47
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k7
r6
ras
relc
rh
rt
Overall Metrics
0.620
0.728
0.620
0.663
Accuracy
Precision
Recall
F1

GEMINI: FEW

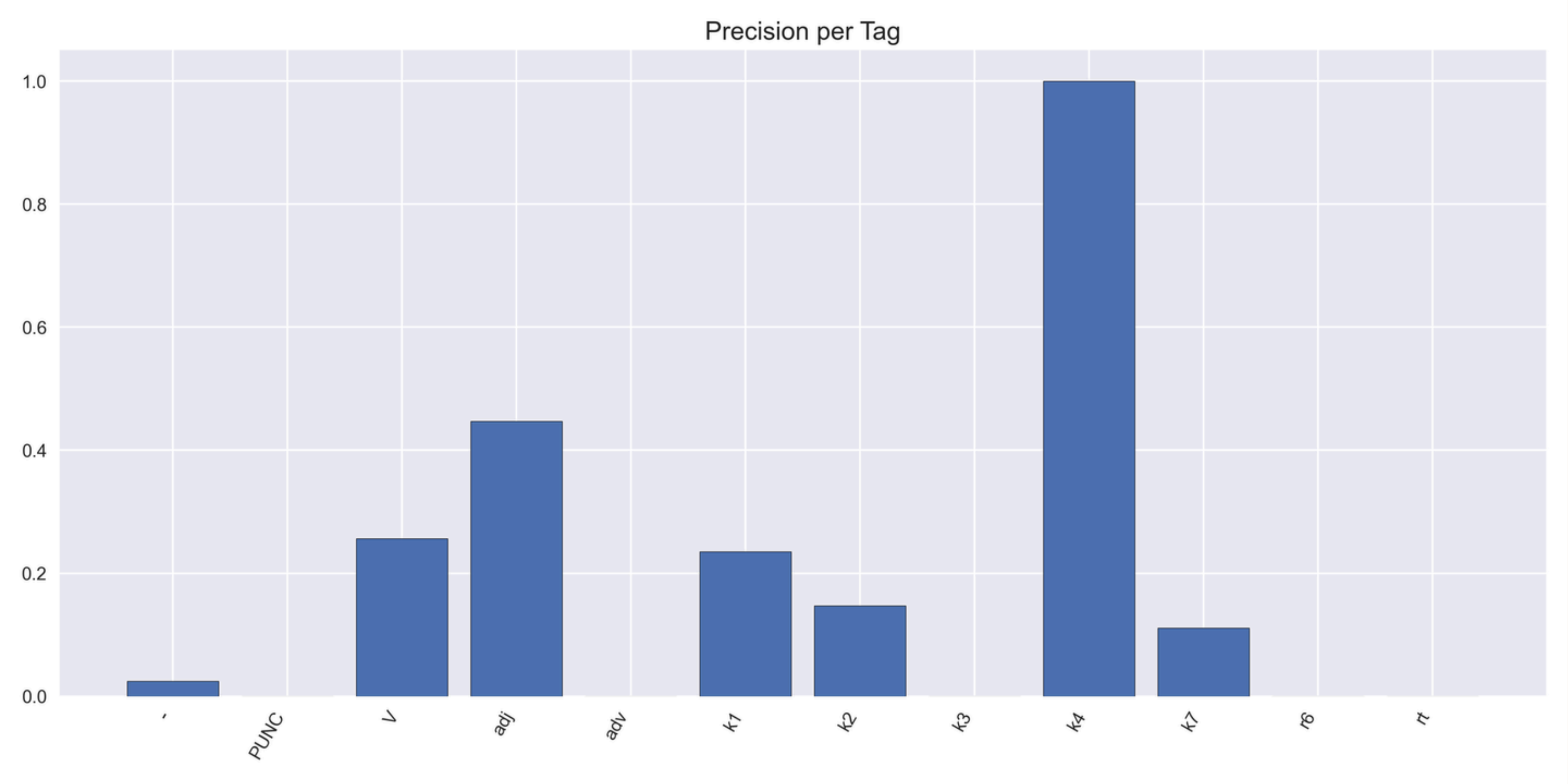

Confusion Matrix: Few Shot Annotation
True label
Predicted label
- PUNC V adj adv k1 k2 k3 k4 k7 r6 rt
- 1 0 8 13 0 2 7 0 0 1 0 0
PUNC 0 0 0 0 0 0 0 0 0 0 0 0
V 12 0 19 20 6 14 6 0 0 1 0 2
adj 15 0 22 76 3 22 12 0 0 2 1 5
adv 1 0 0 3 0 0 2 0 0 0 0 0
k1 4 0 17 21 0 12 1 0 0 0 0 1
k2 4 1 6 30 1 1 5 0 0 4 3 1
k3 1 0 0 1 0 0 0 0 0 0 0 0
k4 1 0 1 4 0 0 1 0 1 0 0 1
k7 2 1 1 2 0 0 0 0 0 1 1 1
r6 0 0 0 0 0 0 0 0 0 0 0 0
rt 0 0 0 0 0 0 0 0 0 0 0 0
0 10 20 30 40 50 60 70



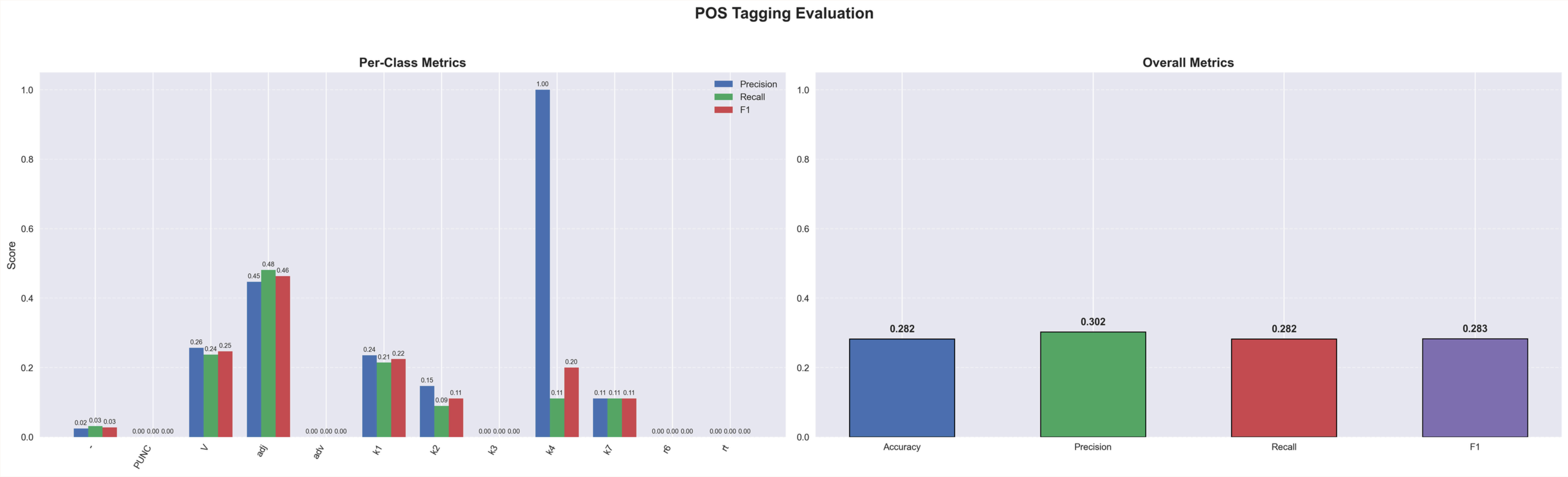
F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt



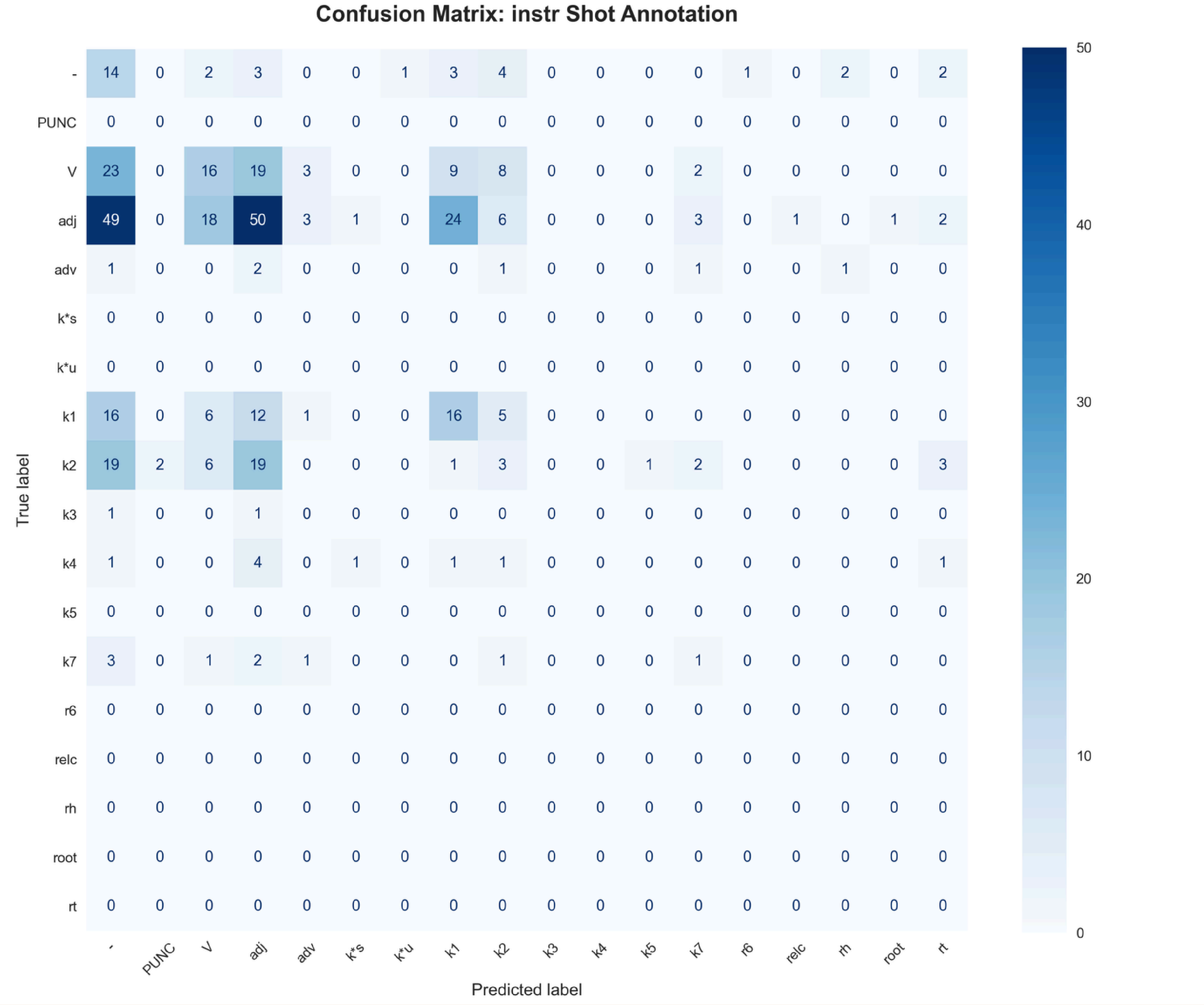
Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt

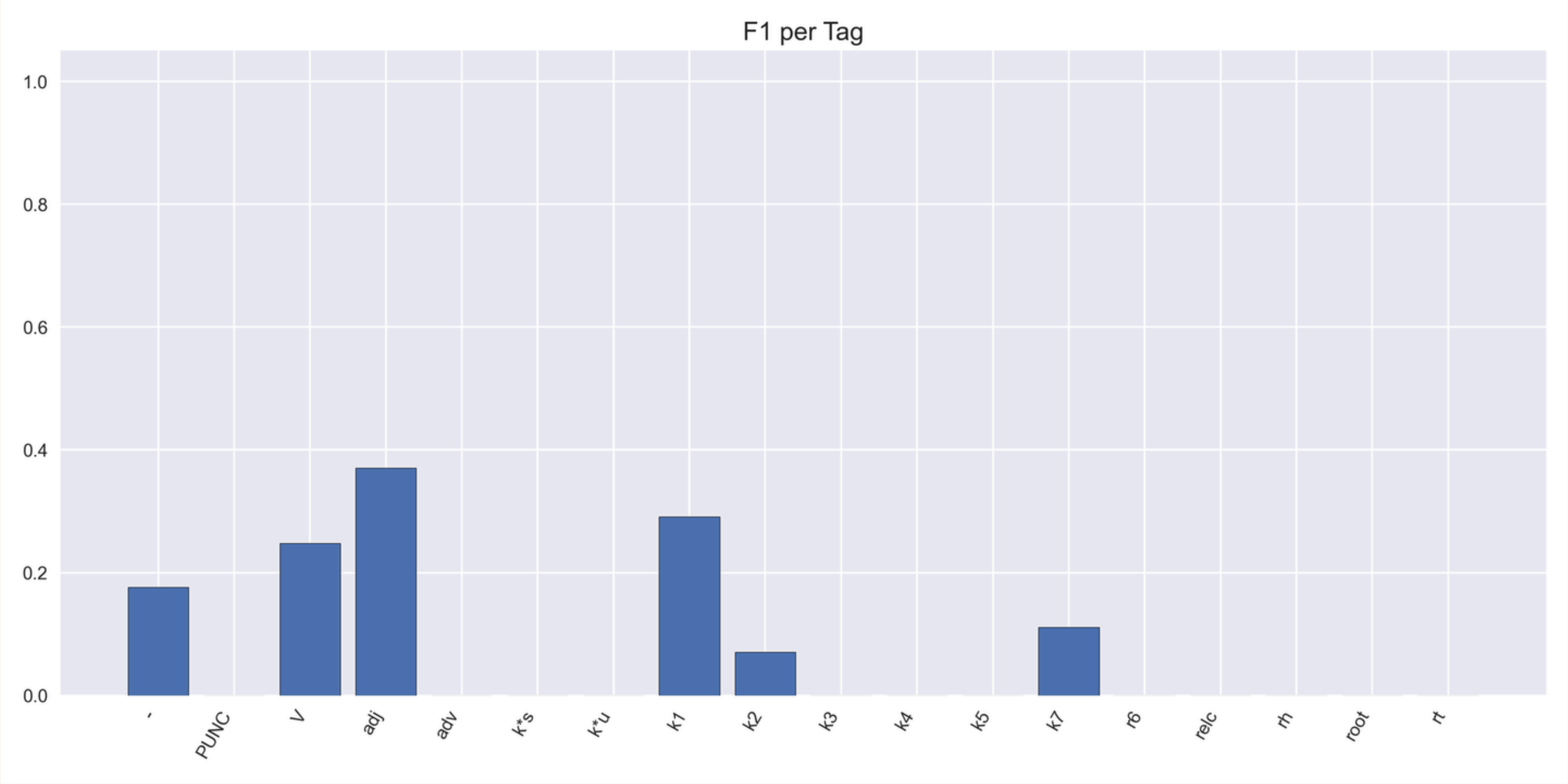


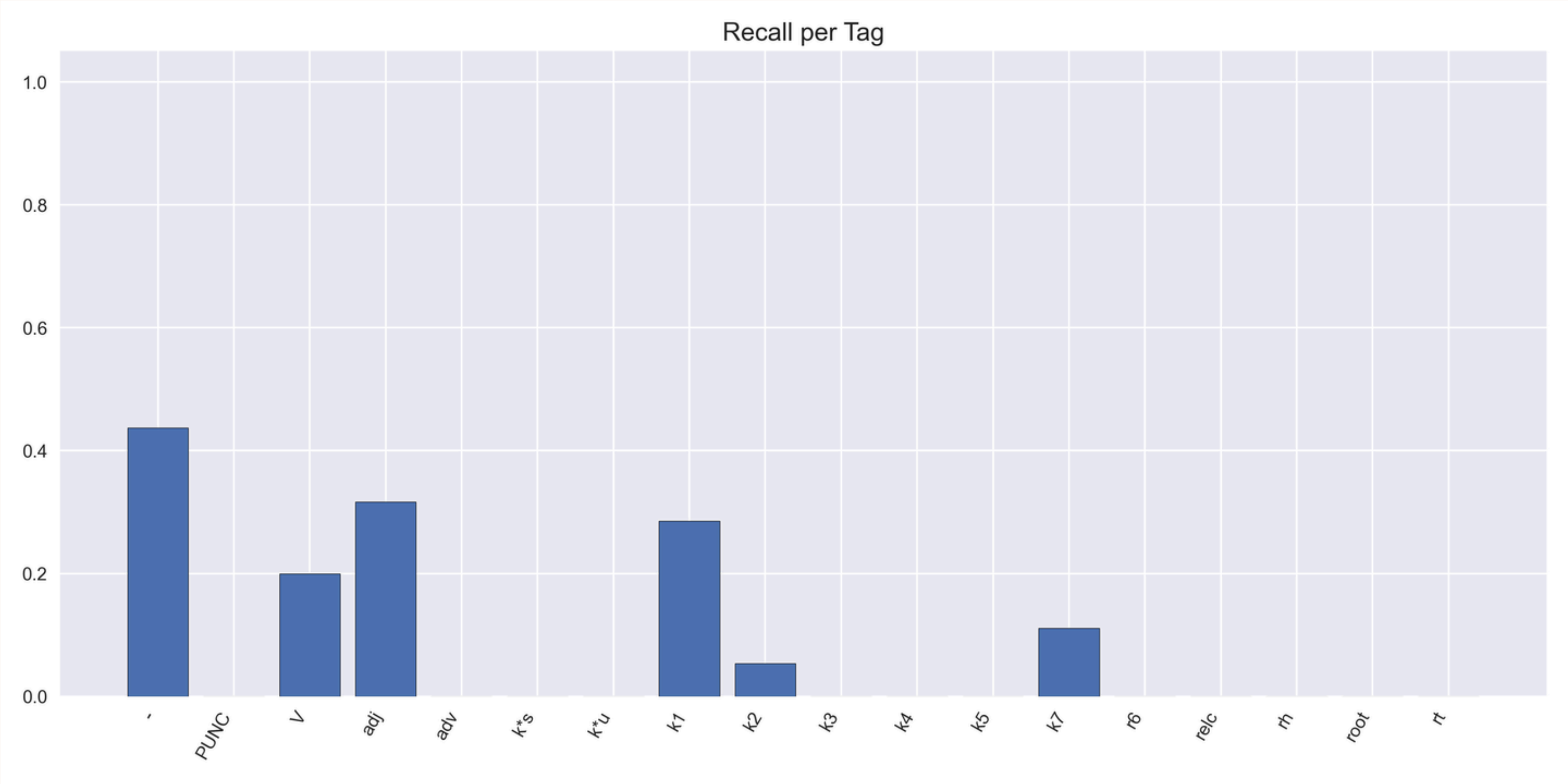
Precision per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt

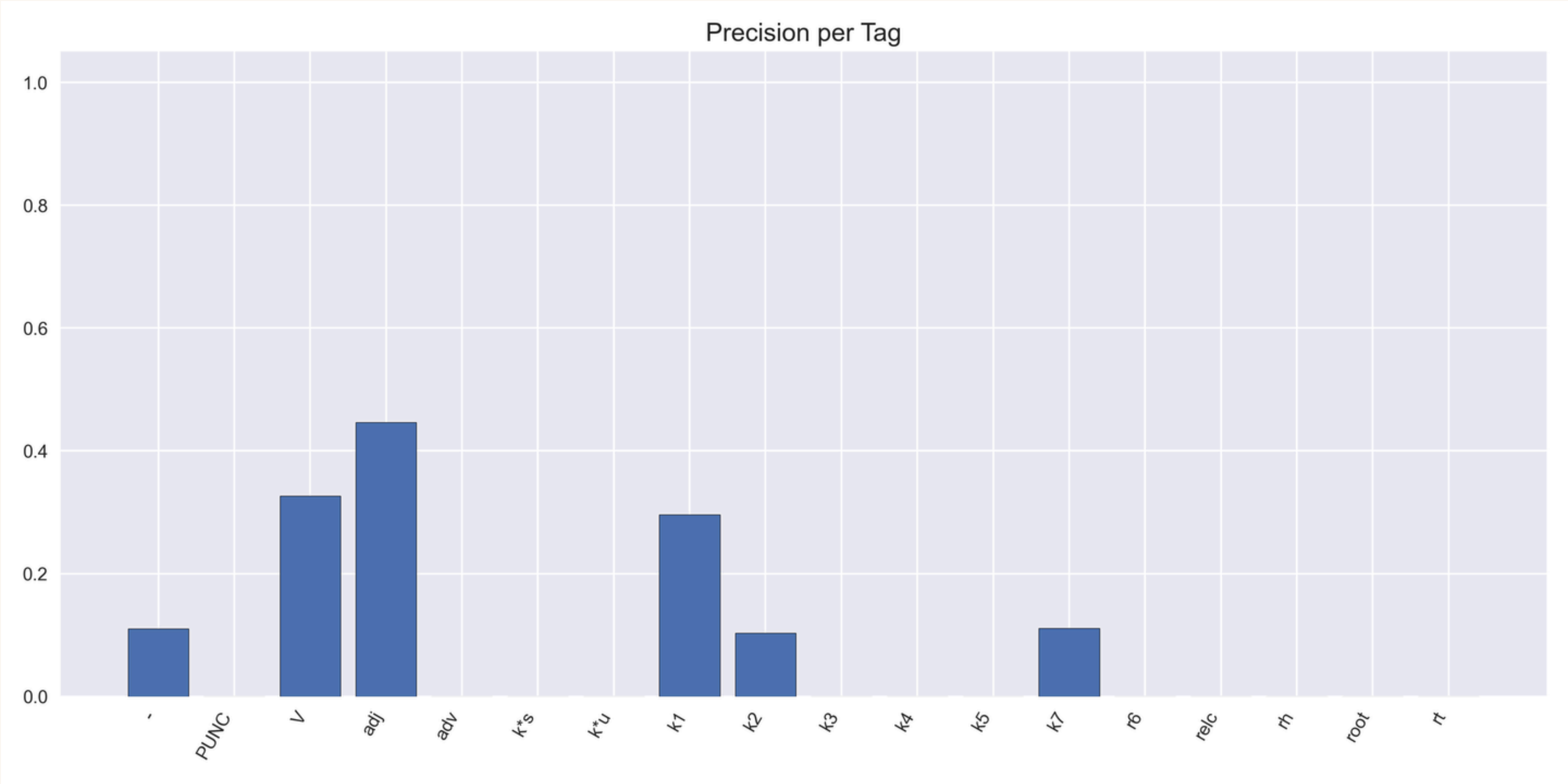

POS Tagging Evaluation
Per-Class Metrics
Overall Metrics
Precision
Recall
F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.02 0.03 0.03
0.00 0.00 0.00
0.26 0.24 0.25
0.45 0.48 0.46
0.00 0.00 0.00
0.24 0.21 0.22
0.15 0.09 0.11
0.00 0.00 0.00
1.00 0.11 0.20
0.11 0.11 0.11
0.00 0.00 0.00
0.00 0.00 0.00
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt
0.282
0.302
0.282
0.283
Accuracy
Precision
Recall
F1

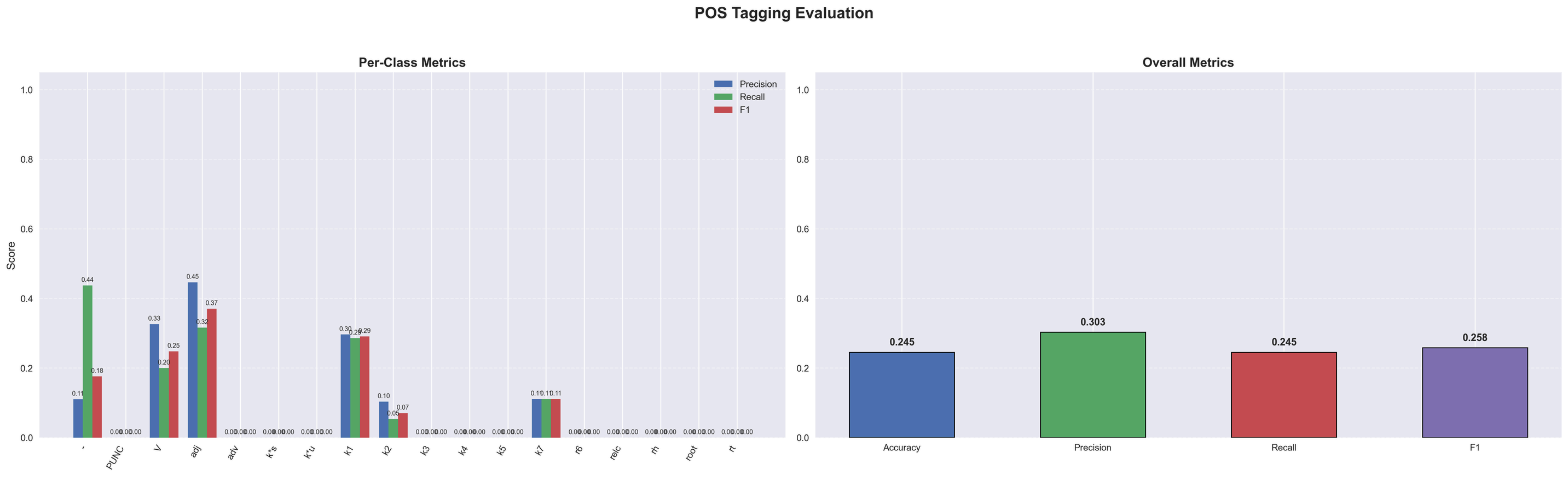
GEMINI: INSTR


Confusion Matrix: instr Shot Annotation
- 14 0 2 3 0 0 1 3 4 0 0 0 0 1 0 2 0 2
PUNC 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
V 23 0 16 19 3 0 0 9 8 0 0 0 2 0 0 0 0 0
adj 49 0 18 50 3 1 0 24 6 0 0 0 3 0 1 0 1 2
adv 1 0 0 2 0 0 0 0 1 0 0 0 1 0 0 1 0 0
k*s 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k*u 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k1 16 0 6 12 1 0 0 16 5 0 0 0 0 0 0 0 0 0
k2 19 2 6 19 0 0 0 1 3 0 0 1 2 0 0 0 0 3
k3 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k4 1 0 0 4 0 1 0 1 1 0 0 0 0 0 0 0 0 1
k5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k7 3 0 1 2 1 0 0 0 1 0 0 0 1 0 0 0 0 0
r6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
relc 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
rh 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
root 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
rt 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
True label
- PUNC V adj adv k*s k*u k1 k2 k3 k4 k5 k7 r6 relc rh root rt
Predicted label
50
40
30
20
10
0

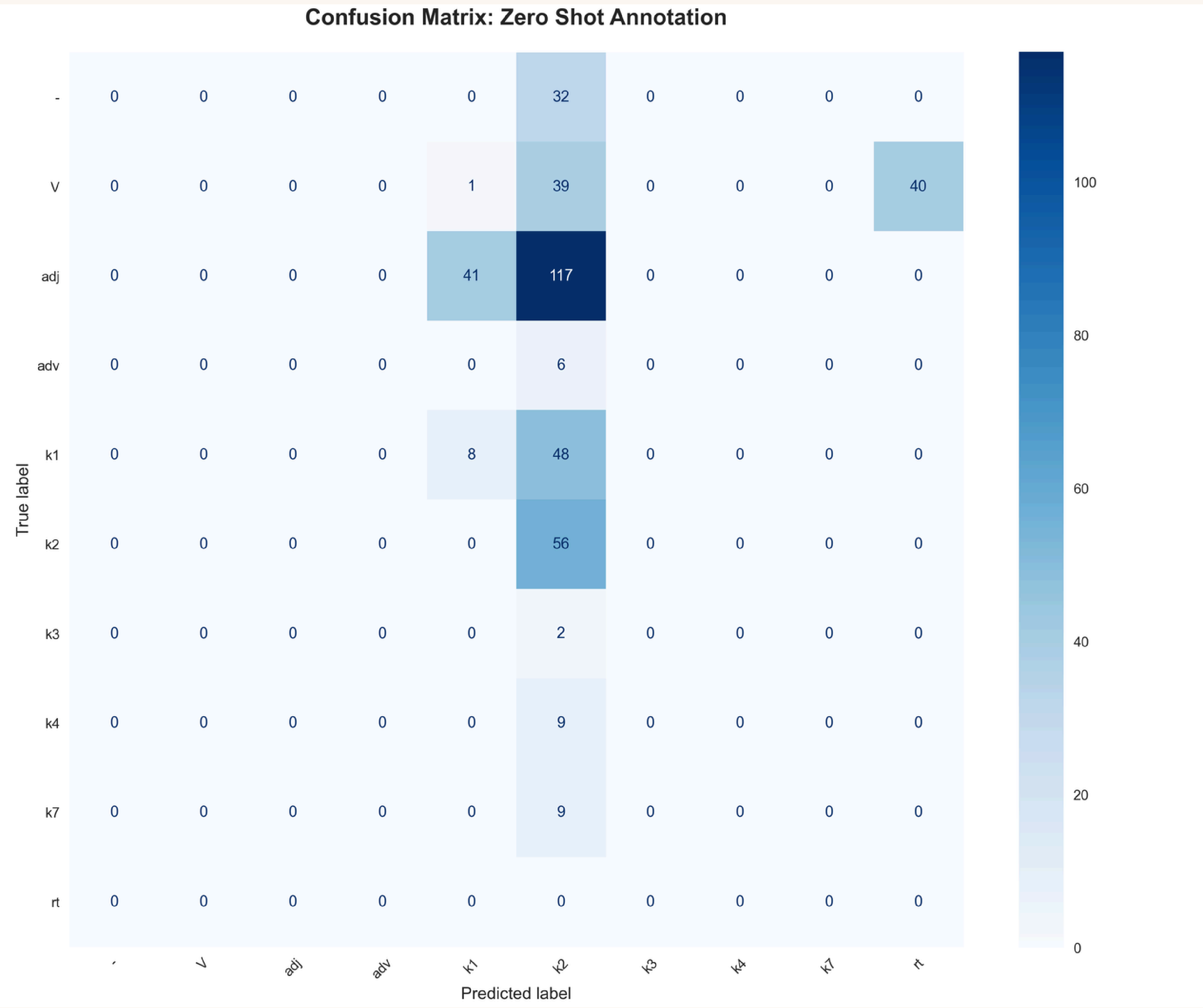

F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
relc
rh
root
rt


Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
relc
rh
root
rt

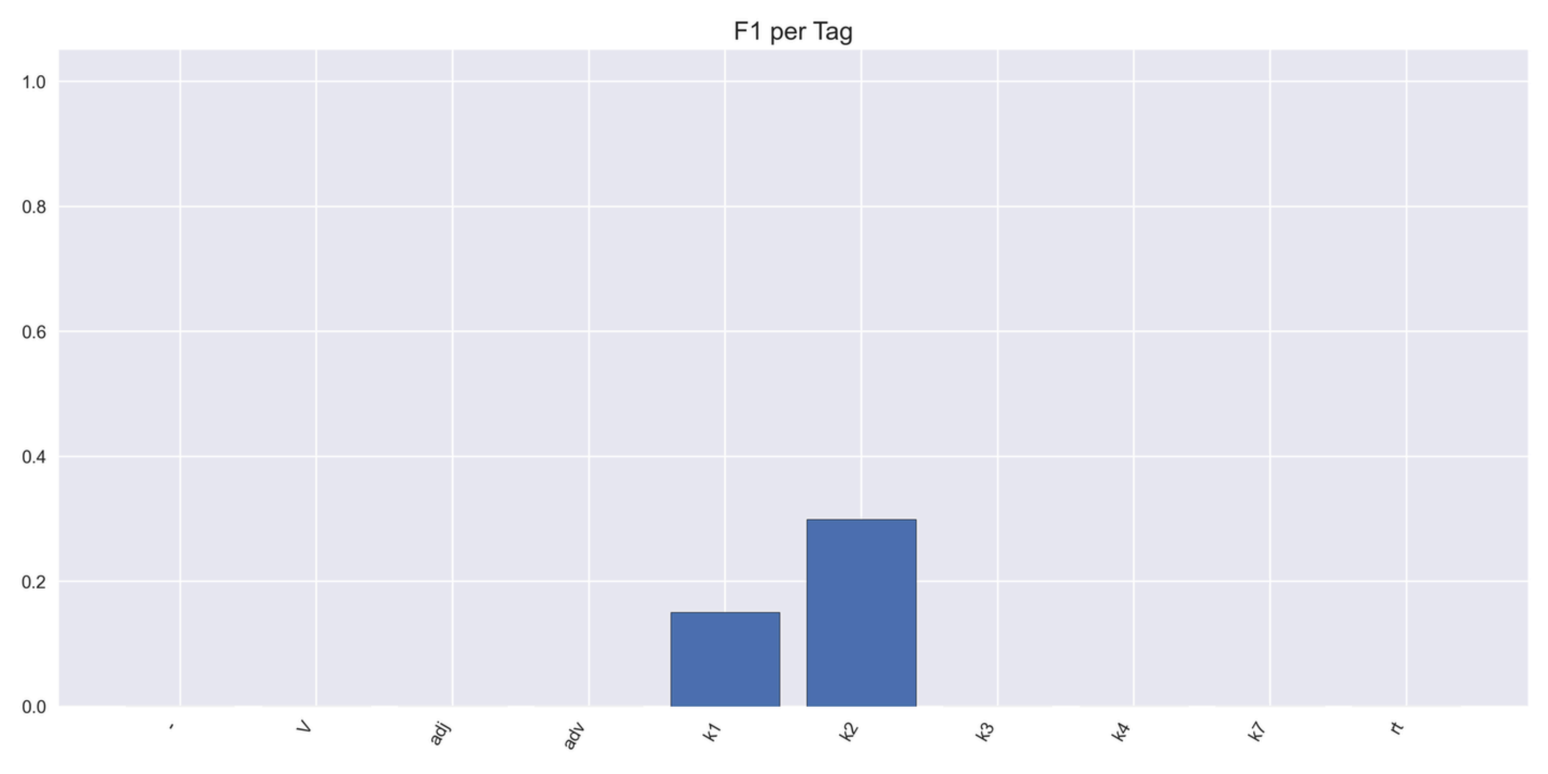

Precision per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
relc
rh
root
rt

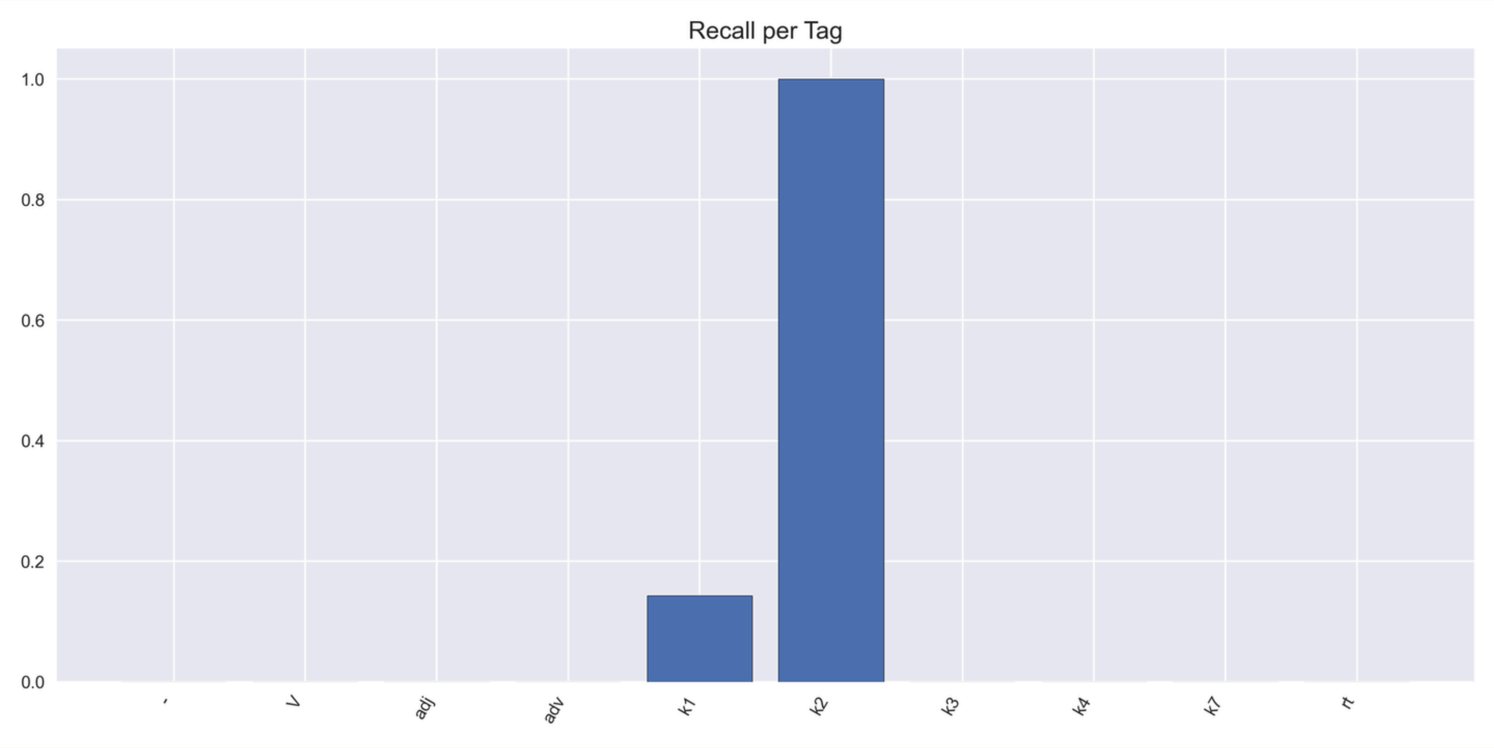

POS Tagging Evaluation
Per-Class Metrics
Precision
Recall
F1
Score
-
PUNC
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
relc
rh
root
rt
Overall Metrics
0.245
0.303
0.245
0.258
Accuracy
Precision
Recall
F1

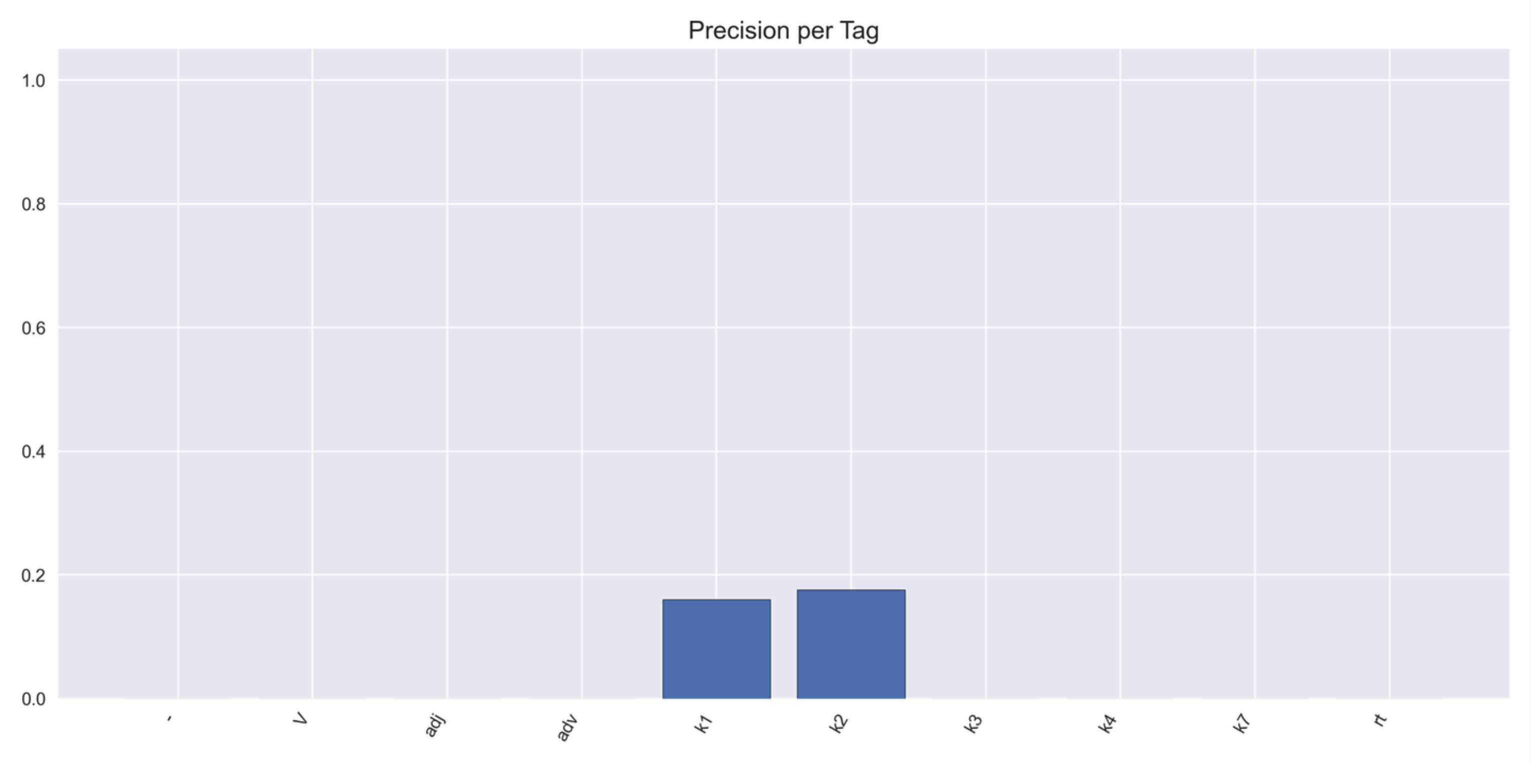
GPT: ZERO

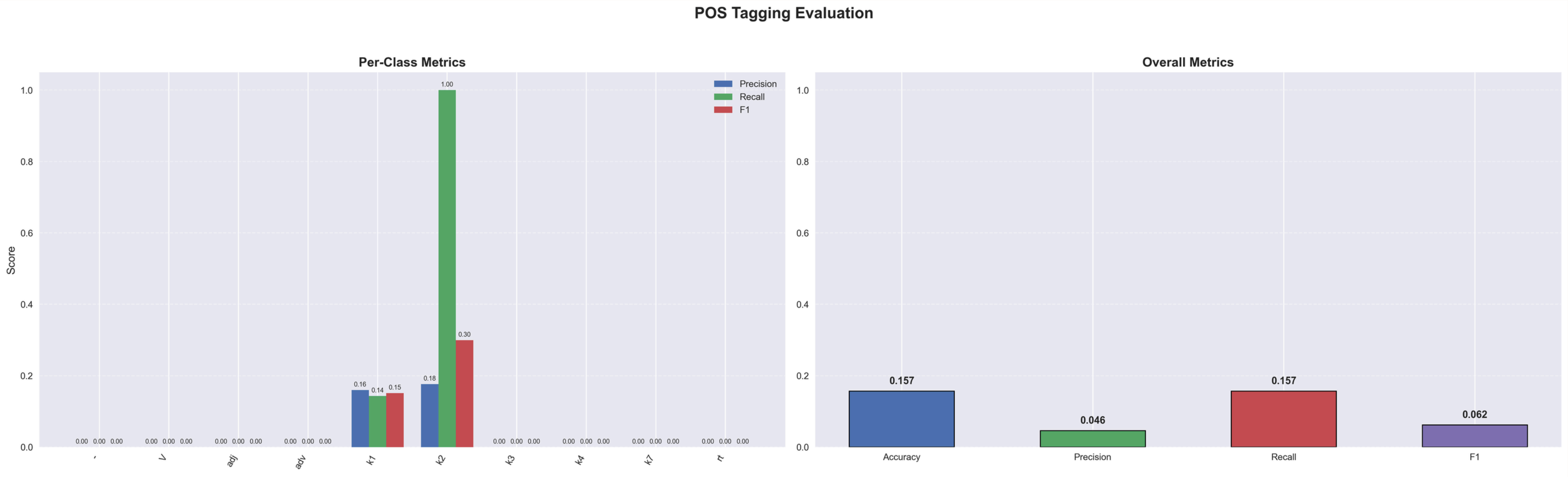


Confusion Matrix: Zero Shot Annotation
True label
Predicted label
- V adj adv k1 k2 k3 k4 k7 rt
- 0 0 0 0 0 32 0 0 0 0
V 0 0 0 0 1 39 0 0 0 40
adj 0 0 0 0 41 117 0 0 0 0
adv 0 0 0 0 0 6 0 0 0 0
k1 0 0 0 0 8 48 0 0 0 0
k2 0 0 0 0 0 56 0 0 0 0
k3 0 0 0 0 0 2 0 0 0 0
k4 0 0 0 0 0 9 0 0 0 0
k7 0 0 0 0 0 9 0 0 0 0
rt 0 0 0 0 0 0 0 0 0 0
100
80
60
40
20
0

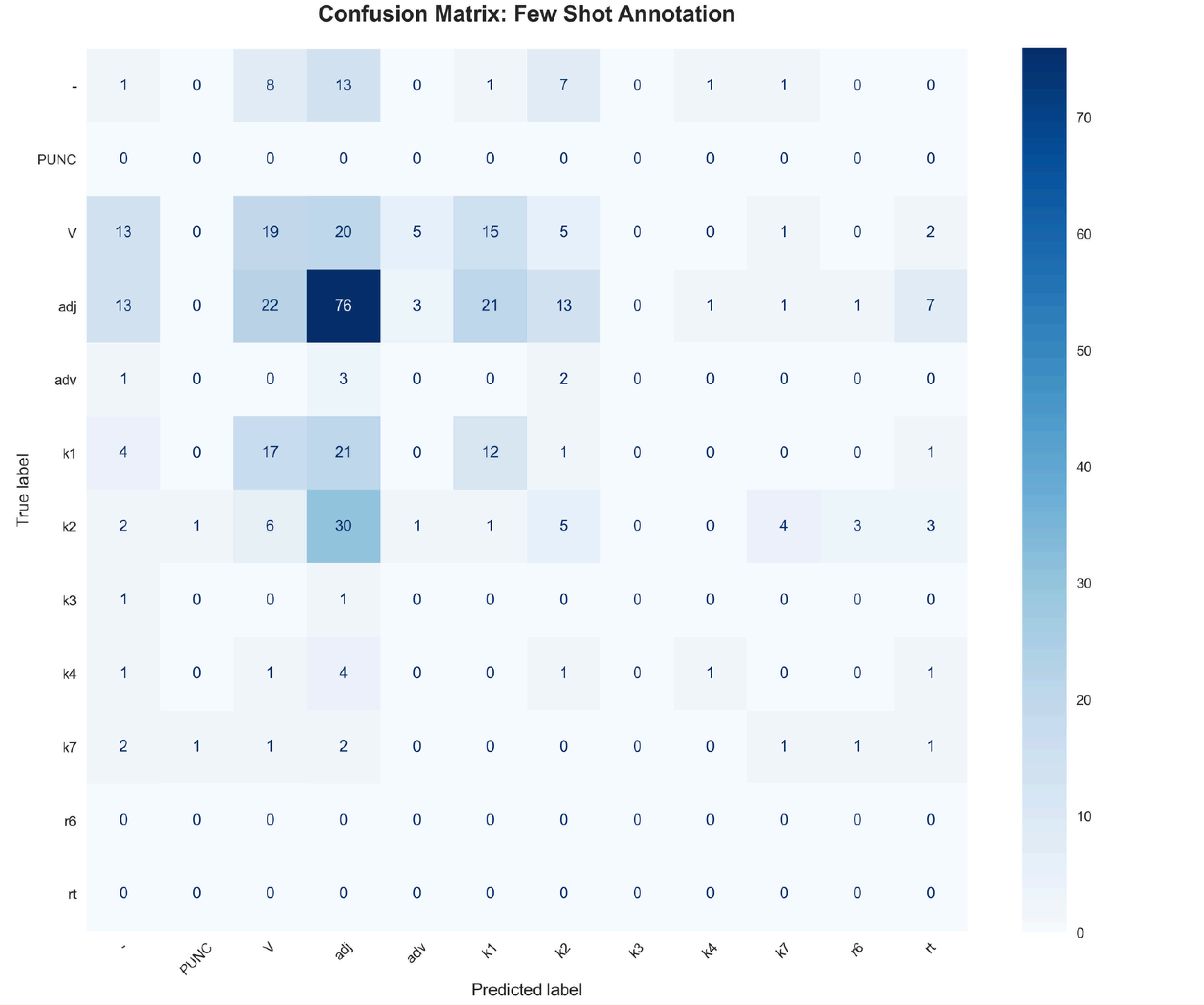

F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k1
k2
k3
k4
k7
rt

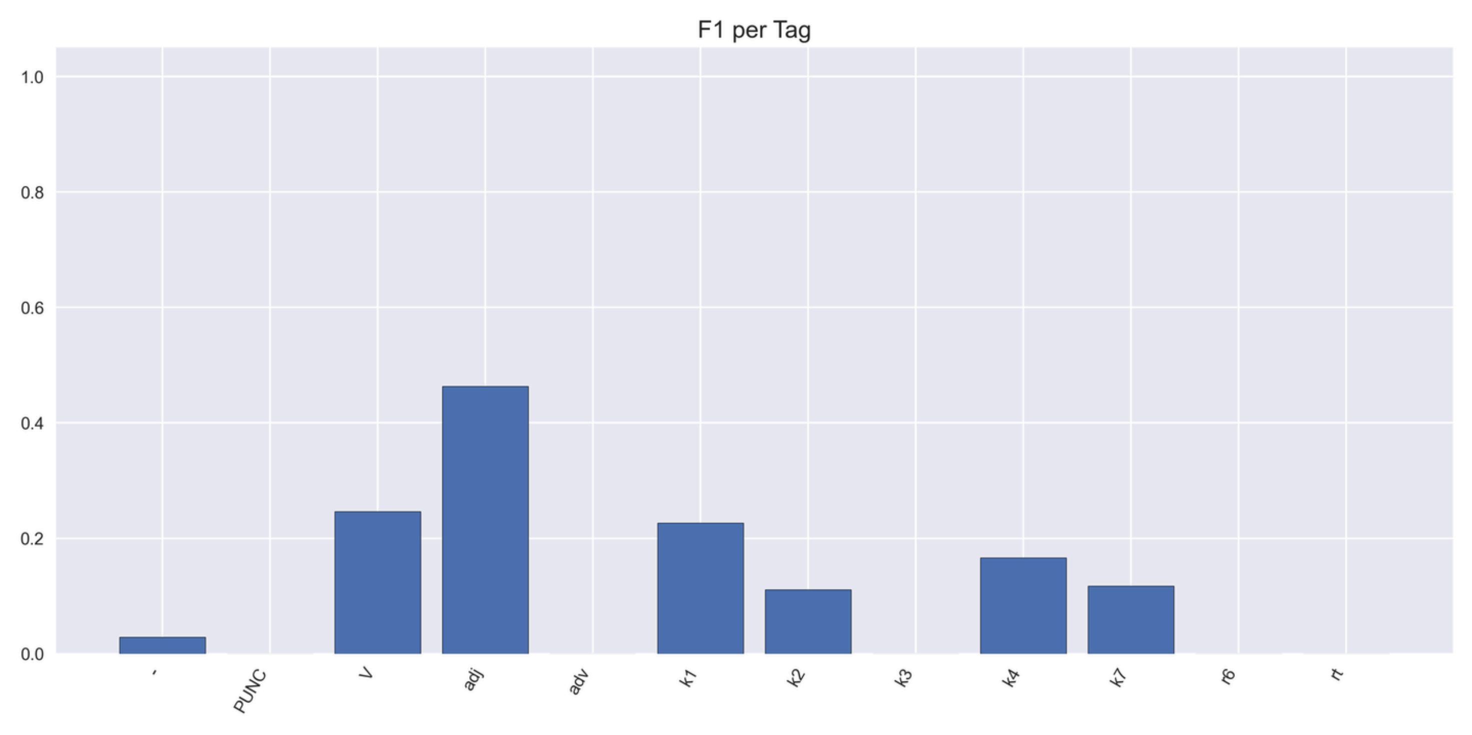

Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k1
k2
k3
k4
k7
rt

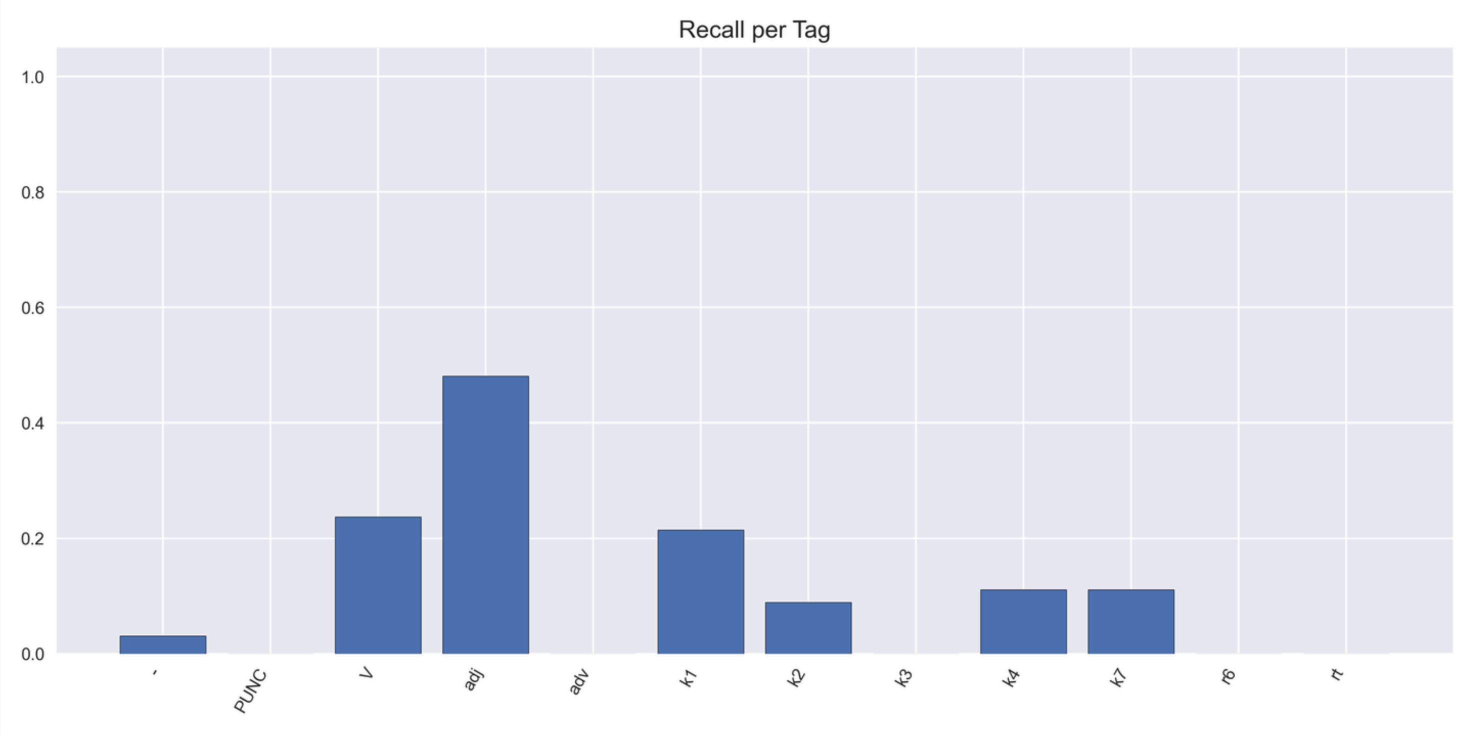

Precision per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k1
k2
k3
k4
k7
rt

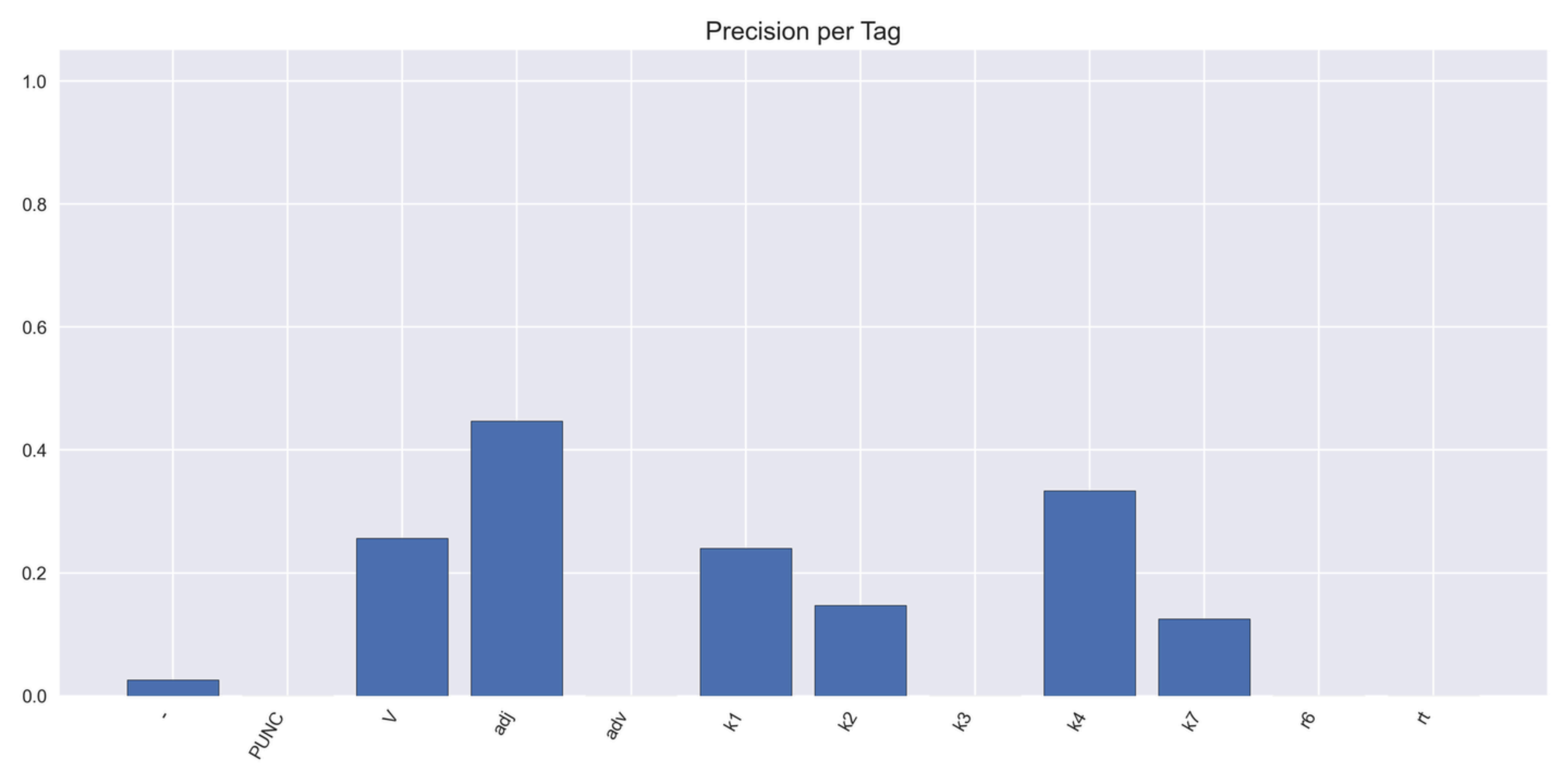

POS Tagging Evaluation
Per-Class Metrics
Precision
Recall
F1
Score
-
V
adj
adv
k1
k2
k3
k4
k7
rt
0.16
0.14
0.15
0.18
1.00
0.30
Overall Metrics
Accuracy
Precision
Recall
F1
0.157
0.046
0.157
0.062

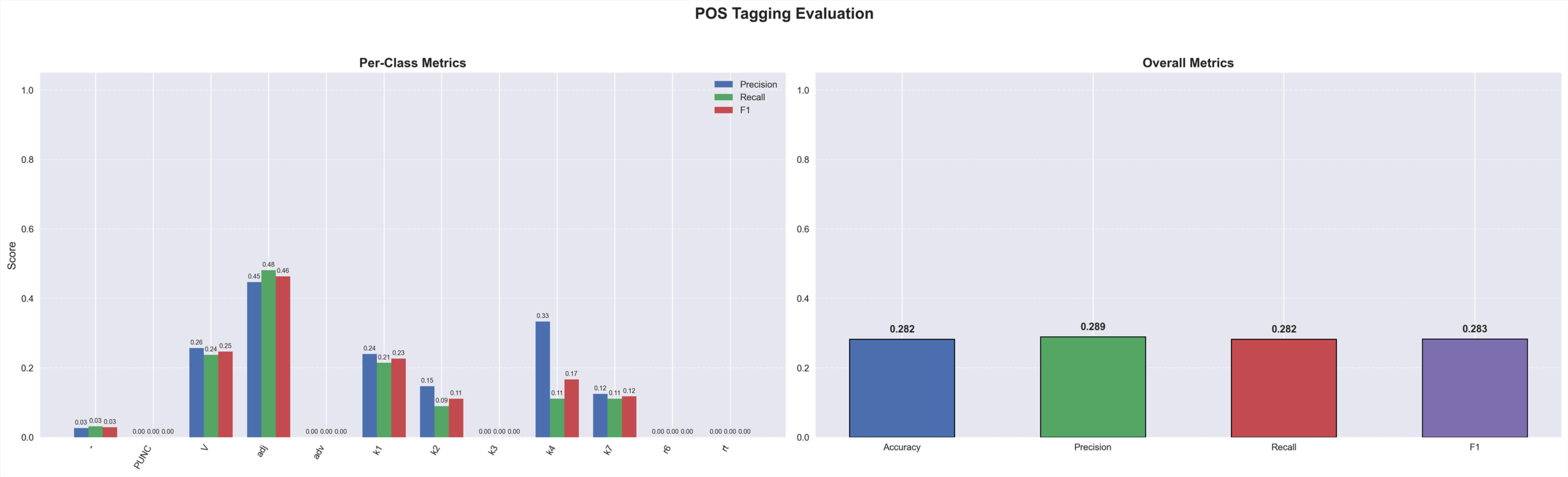
GPT: FEW


Confusion Matrix: Few Shot Annotation
True label
Predicted label
- PUNC V adj adv k1 k2 k3 k4 k7 r6 rt
- 1 0 8 13 0 1 7 0 1 1 0 0
PUNC 0 0 0 0 0 0 0 0 0 0 0 0
V 13 0 19 20 5 15 5 0 0 1 0 2
adj 13 0 22 76 3 21 13 0 1 1 1 7
adv 1 0 0 3 0 0 2 0 0 0 0 0
k1 4 0 17 21 0 12 1 0 0 0 0 1
k2 2 1 6 30 1 1 5 0 0 4 3 3
k3 1 0 0 1 0 0 0 0 0 0 0 0
k4 1 0 1 4 0 0 1 0 1 0 0 1
k7 2 1 1 2 0 0 0 0 0 1 1 1
r6 0 0 0 0 0 0 0 0 0 0 0 0
rt 0 0 0 0 0 0 0 0 0 0 0 0
70 60 50 40 30 20 10 0


F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt

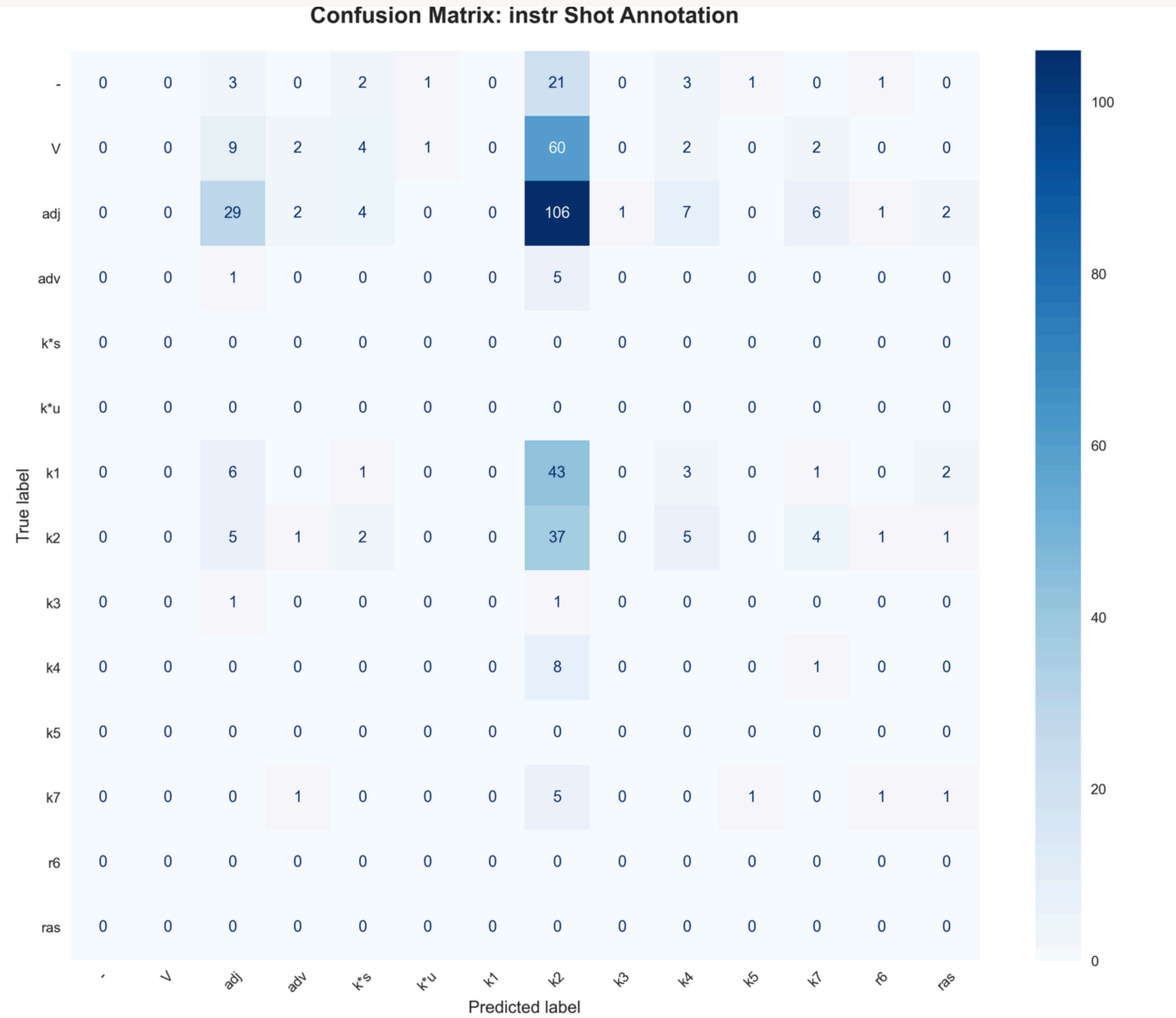

Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt


Precision per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt

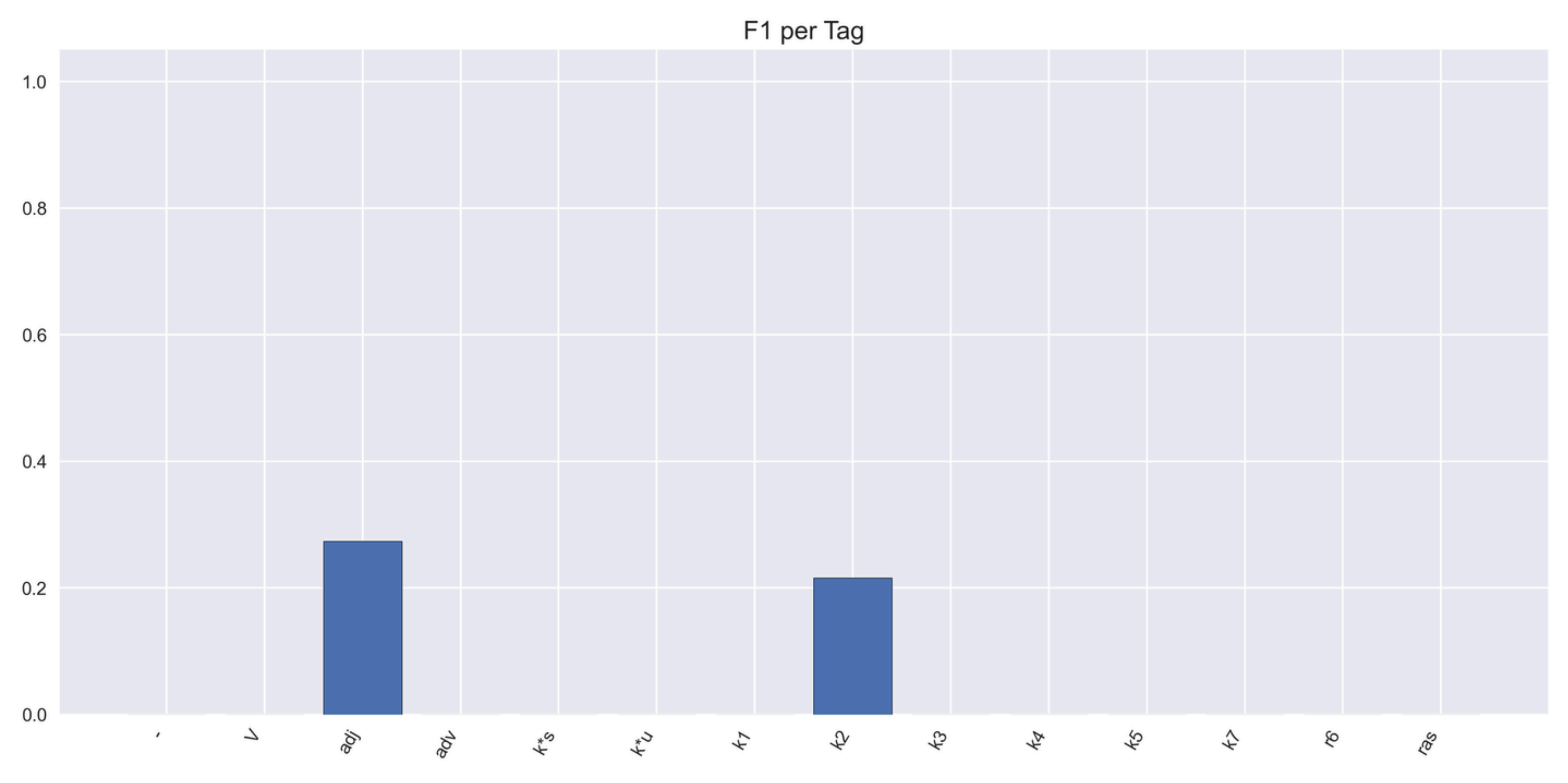


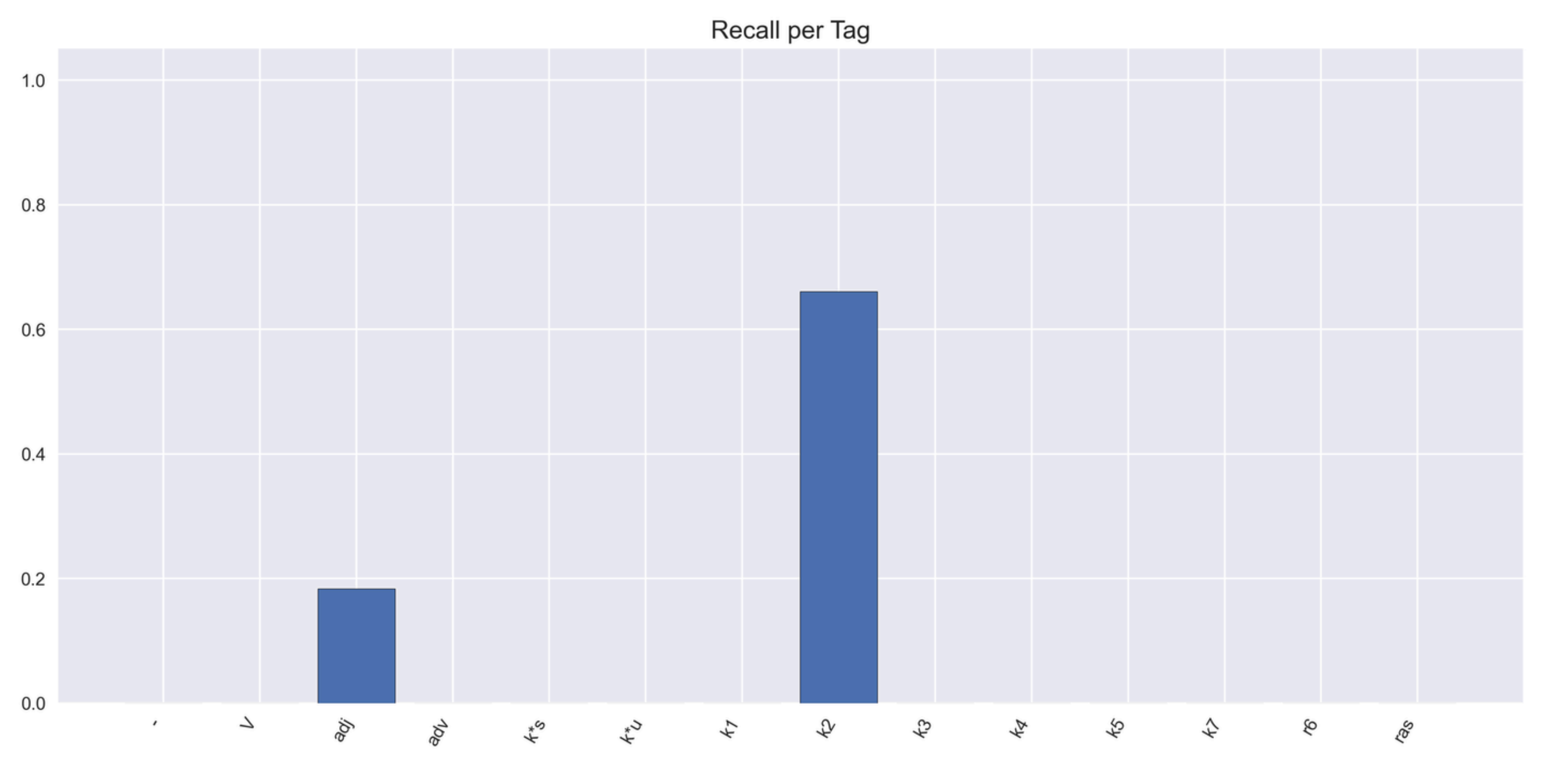
POS Tagging Evaluation
Per-Class Metrics
Precision
Recall
F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.03 0.03 0.03
0.00 0.00 0.00
0.26 0.24 0.25
0.45 0.48 0.46
0.00 0.00 0.00
0.24 0.21 0.23
0.15 0.09 0.11
0.00 0.00 0.00
0.33 0.11 0.17
0.12 0.11 0.12
0.00 0.00 0.00
0.00 0.00 0.00
-
PUNC
V
adj
adv
k1
k2
k3
k4
k7
r6
rt
Overall Metrics
0.282
0.289
0.282
0.283
Accuracy
Precision
Recall
F1

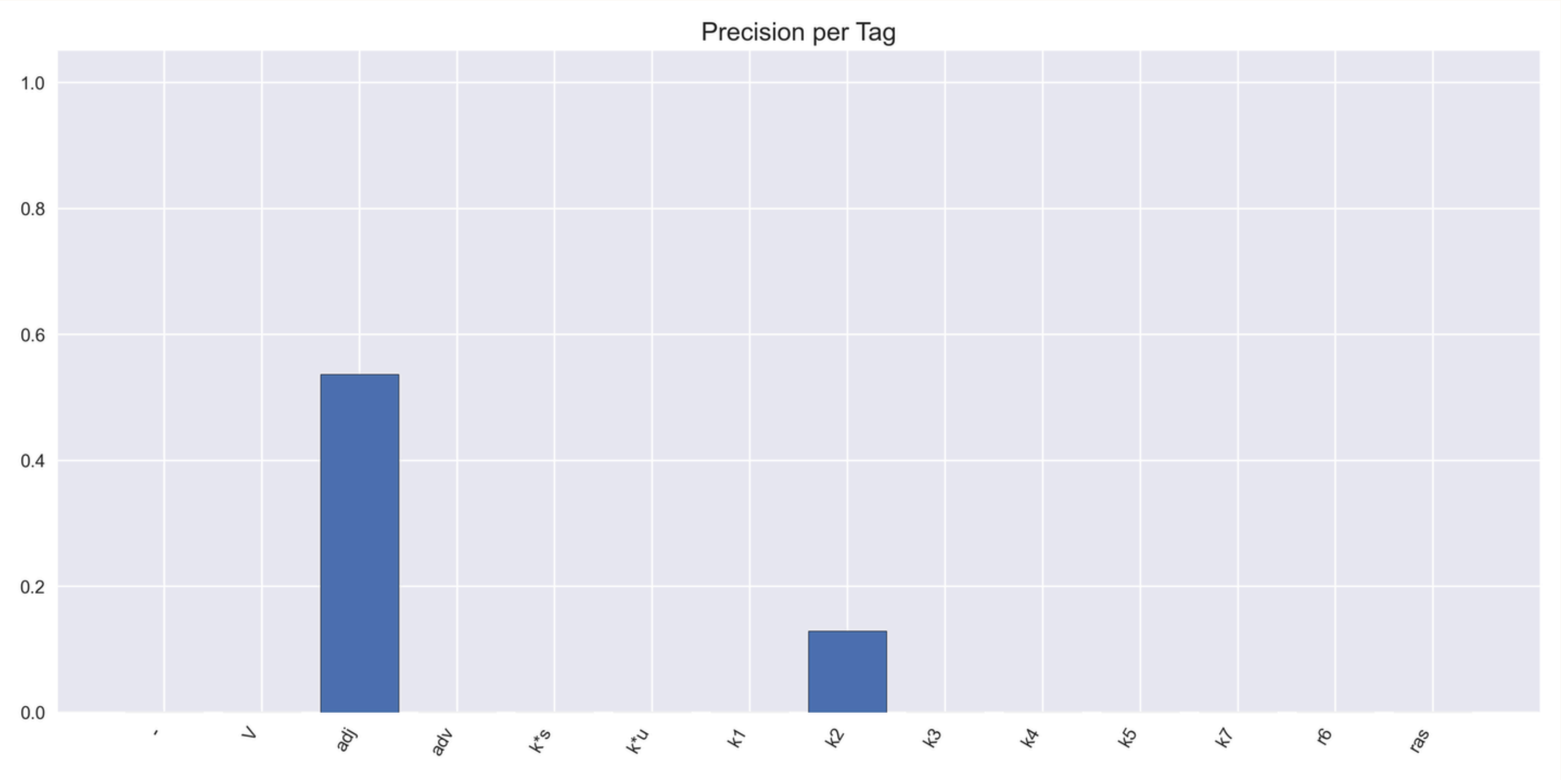
GPT: INSTR

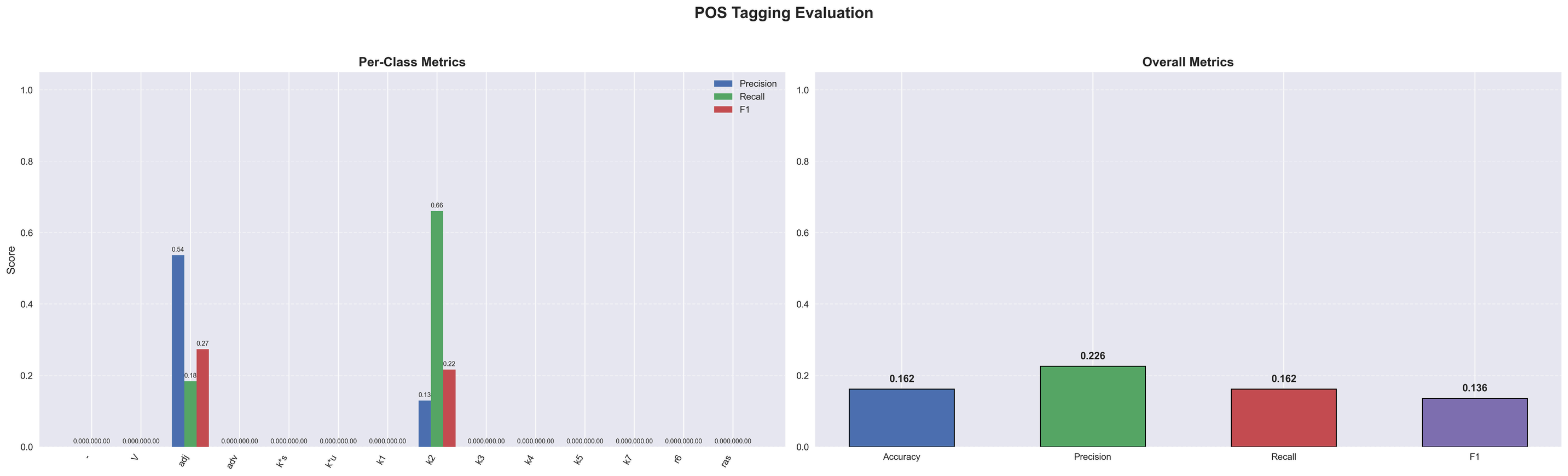


Confusion Matrix: instr Shot Annotation
True label
Predicted label
- V adj adv k*s k*u k1 k2 k3 k4 k5 k7 r6 ras
- 0 0 3 0 2 1 0 21 0 3 1 0 1 0
V 0 0 9 2 4 1 0 60 0 2 0 2 0 0
adj 0 0 29 2 4 0 0 106 1 7 0 6 1 2
adv 0 0 1 0 0 0 0 5 0 0 0 0 0 0
k*s 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k*u 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k1 0 0 6 0 1 0 0 43 0 3 0 1 0 2
k2 0 0 5 1 2 0 0 37 0 5 0 4 1 1
k3 0 0 1 0 0 0 0 1 0 0 0 0 0 0
k4 0 0 0 0 0 0 0 8 0 0 0 1 0 0
k5 0 0 0 0 0 0 0 0 0 0 0 0 0 0
k7 0 0 0 1 0 0 0 5 0 0 1 0 1 1
r6 0 0 0 0 0 0 0 0 0 0 0 0 0 0
ras 0 0 0 0 0 0 0 0 0 0 0 0 0 0
100
80
60
40
20
0


F1 per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
ras


Recall per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
ras


Precision per Tag
1.0
0.8
0.6
0.4
0.2
0.0
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
ras


POS Tagging Evaluation
Per-Class Metrics
Precision
Recall
F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.54
0.18
0.27
0.66
0.13
0.22
0.000.000.00
-
V
adj
adv
k*s
k*u
k1
k2
k3
k4
k5
k7
r6
ras
Overall Metrics
0.162
0.226
0.162
0.136
Accuracy
Precision
Recall
F1

CROSS LINGUISTIC

```
# Sandesh
# Bharat ane Pakistan vache yuddhani sthiti sarjai.
1   Bharat - India
2   ane - and
3   Pakistan - Pakistan
4   vache - between
5   yuddhani - of war
6   sthiti - situation
7   sarjai - created/arose
8   . - punc
```

```
# Divya Bhaskar
# Rajkotma nava airportnu lokarpan PM Modi karshe.
1   Rajkotma - in Rajkot
2   nava - new
3   airportnu - of the airport
4   lokarpan - inauguration
5   PM - PM
6   Modi - Modi
7   karshe - will do.fut
8   . - punc
```

```
# Andhra Jyothi
# Kendram budget guttu vippina PM Modi.
1   Kendram - center/union
2   budget - budget
3   guttu - secret
4   vippina - opened/revealed
5   PM - PM
6   Modi - Modi
7   . - punc
```

```
# Eenadu News
# Prapanchamlone tholi air taxi station Dubai lo siddham ayyindi.
1   Prapanchamlone - in the world itself
2   tholi - first
3   air - air
4   taxi - taxi
5   station - station
6   Dubai - Dubai
7   lo - in
8   siddham - ready
9   ayyindi - became.pst
10  . - punc
```

Gujarati data

dhabakedar - heavy/vigorous; is slang

lokarpan - inauguration ; too polished/authentic, prefer to use english word

hirabaajarma - in the diamond market; brand name now used as synonymous

sachivalayanu - of the secretariat; too polished/authentic, prefer to use english word

rahi - stayed.pst; rahi is pure pronunciation, ppl just say rai

aanchka - tremors; too authentic/formal, pure, not used

khelaiyao - professional players; used to refer to ppl very good at garba (ontext dependent)

ghabrakat - panic/fear; not used frequently

dharatmate - for the land; uncommon, too pure

hethal - under; uncommon, too pure

taukane - could simply use some other simpler word from the Language or use English, preferred

rupiya be - used altogether as a phrase to represent “one-two rupees here or there”

Navinikaran - uncommon, usually replaced with simpler words like renovation

nava akarshano - might be referring to sarcastic way of saying that “new incidents” have happened lately, contextual dependency

Praman - depending on context, it could either mean “quantity (like here)” or “proof”

Thaya - contextually, could mean either “became.pst” or “happened”

Paryatakoni - little too authentic, usually speakers might end up using the English alternative

Lokarpan - uncommon, average speaker might just use English alternative

Malkat - uncommon, avg speaker might use alternatives from other languages with easier understanding

Tabakkanu - advanced speaker might use English alternative mathamarchidi - religious conversion; isn’t really used

chudandi - see.imp; honorifics in see chudu v/s chudandi “andi” is respect

fida - impressed ; telangana regional word

noukalanu - ships.acc; not really used, just use english word ships

eduru gaali- strong wind; metaphor for opposition

sarathi - charioteer; sanskrit word used

sankshobham - scarcity; generally not used difficult word

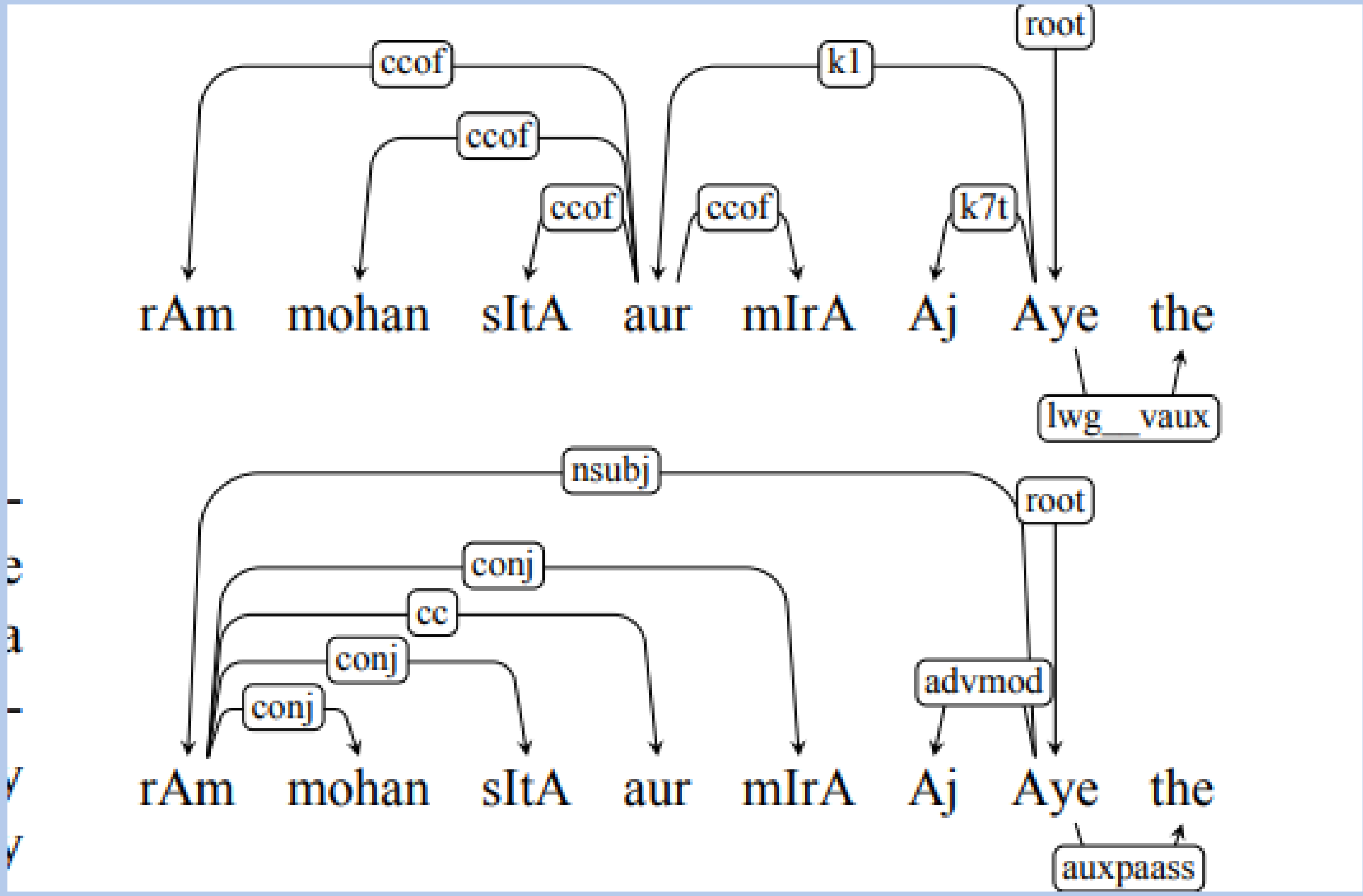
labhadharula - of beneficiaries(used in agricultural terms for immediate buyer of base product) ; not really used in day to day convo

vyangya - satirical; sanskrit word not used in telugu

ulla - onion; just call it english word onion, this is funny

UD CONVERSION

Paninian Grammar vs Universal Dependencies



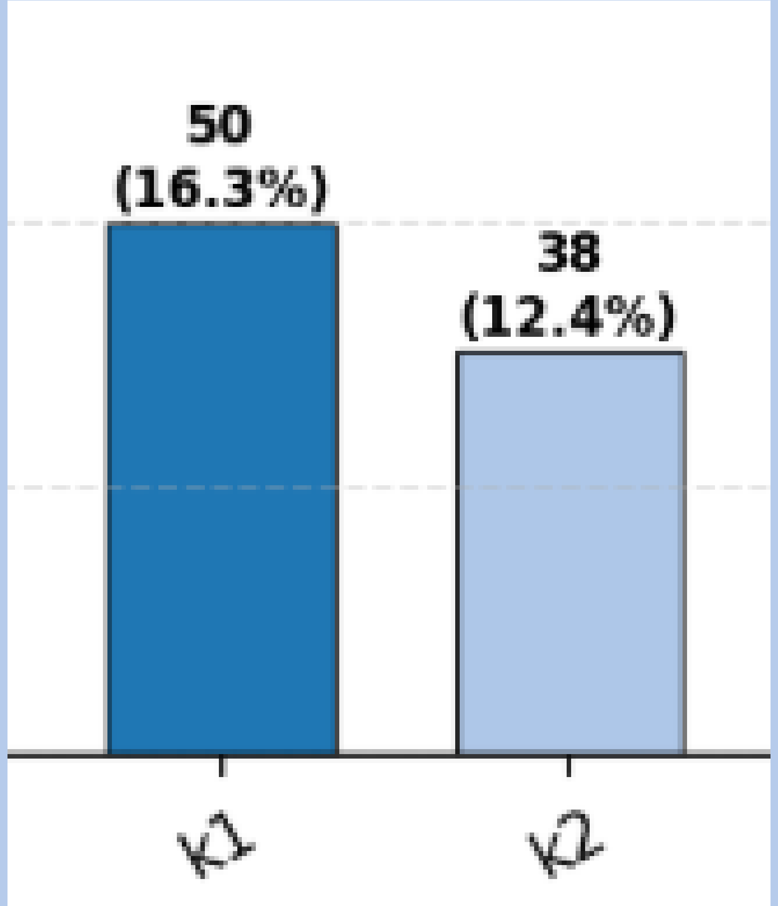
```
# The Hindu
# The government announced new research grants for students.
1   The amod
2   government  nsubj
3   announced   V
4   new amod
5   research    amod
6   grants  dobj
7   for -
8   students    iobj
9   .   PUNC
```

```
# BBC News
# Global temperatures reached a record high last month.
1   Global  amod
2   temperatures    nsubj
3   reached V
4   a    amod
5   record  amod
6   high    amod
7   last    amod
8   month   dobj
9   .   PUNC
```

```
deepti_to_ud = {
    "k1":"nsubj",
    "k2":"dobj",
    "k3":"nmod",
    "k4":"iobj",
    "k5":"nmod",
    "k7":"nmod",
    "rt":"advmod",
    "rh":"advmod",
    "ras":"nmod",
    "k*u":"vocative",
    "k*s":"xcomp",
    "r6":"nmod",
    "relc": "acl",
    "rs":"nmod",
    "adv":"advmod",
    "adj":"amod",
}
```

CONCLUSION

K1 and K2 imbalance



Data we used is sentences is from the news.
Thus, it follows a very predictable pattern of Agent - Verb - Patient.
This diff is even more pronounced for English, as it relies heavily on word order making these roles even more apparent.

Class imbalance Issues:
Models trained on this data will overpredict k1 and k2.
Hence, a varied dataset is essential

English vs Paninian

01 Lack of Case Markers

02 Prepositional Ambiguity

03 Structural Differences

Pipeline

01 Manual Annotation

02 Tagset Modification

03 Automated Annotation

Pipeline

04 Stress Testing AI models

05 Cross Linguistic Annotation

06 UD Conversion

THANK YOU VERY MUCH!